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(54) **POWER TOOL STAND**

(71) Applicant: **BLACK & DECKER INC.**, New Britain, CT (US)

(72) Inventors: **Rafael Gottschling**, Selters-Eisenbach (DE); **Daniel Kadleck**, Bischofsheim (DE); **Birju Patel**, Ruislip (GB)

(73) Assignee: **Black & Decker Inc.**, New Britain, CT (US)

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See application file for complete search history.

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Primary Examiner — Michelle Lopez

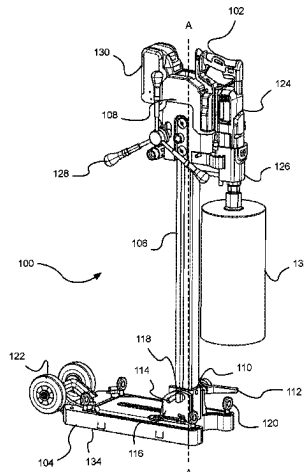
(74) *Attorney, Agent, or Firm* — Womble Bond Dickinson (US) LLP

(57)

ABSTRACT

A power tool stand for a removeable power tool has a first battery interface and a removeable battery having a first tool interface. The power tool stand comprises a base and a projecting frame connected to the base. A power tool carriage is moveably mounted on the projecting frame and the power tool carriage comprising a second battery interface and a second tool interface. The second battery interface is compatible with the first tool interface and the second battery interface is arranged to electrically and mechanically connect together with first tool interface. The second tool interface is compatible with the first battery interface and the second tool interface is arranged to electrically and mechanically connect together with first battery interface. The first battery interface and the first tool interface are incompatible.

20 Claims, 15 Drawing Sheets



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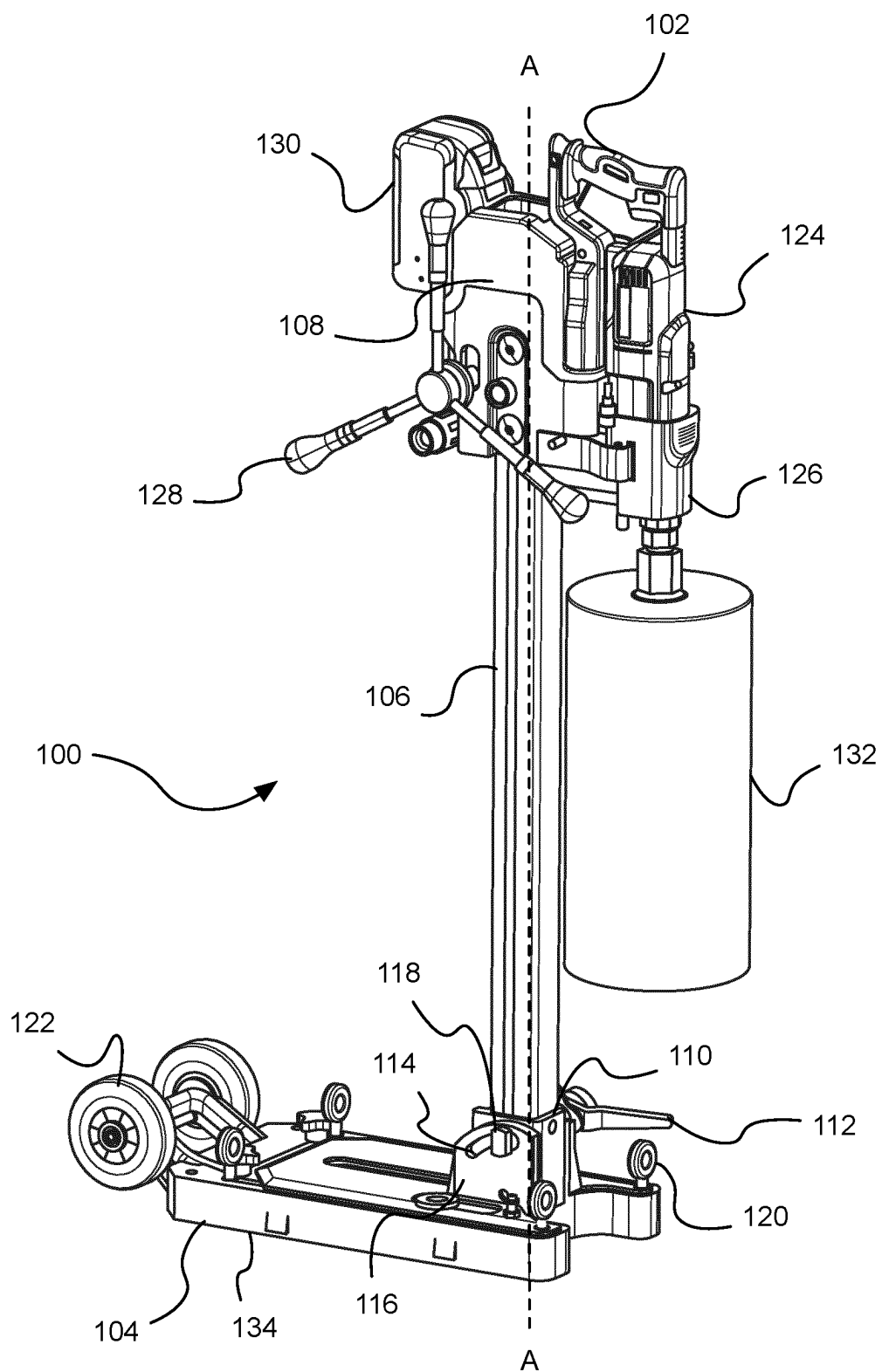


Fig. 1

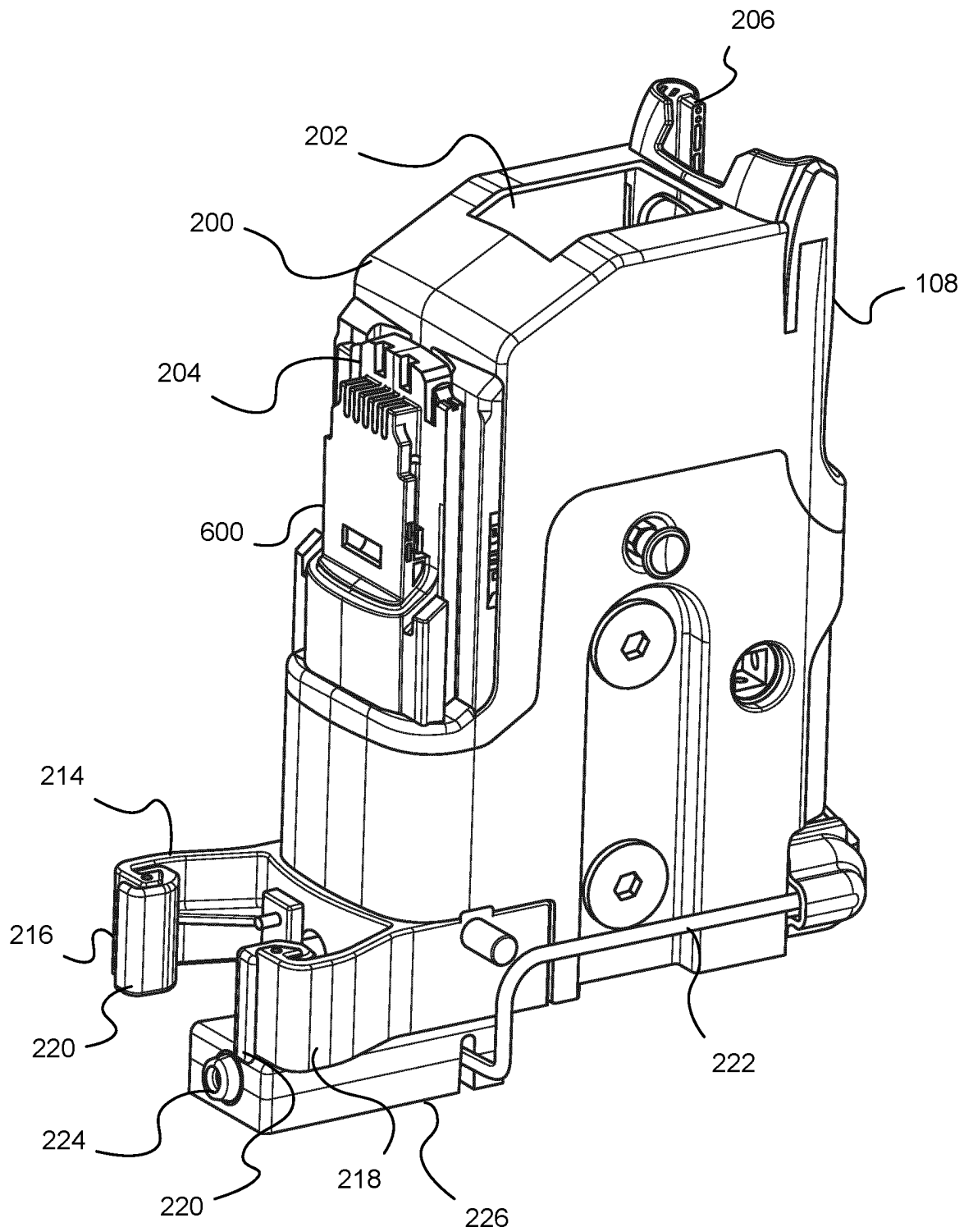


Fig. 2

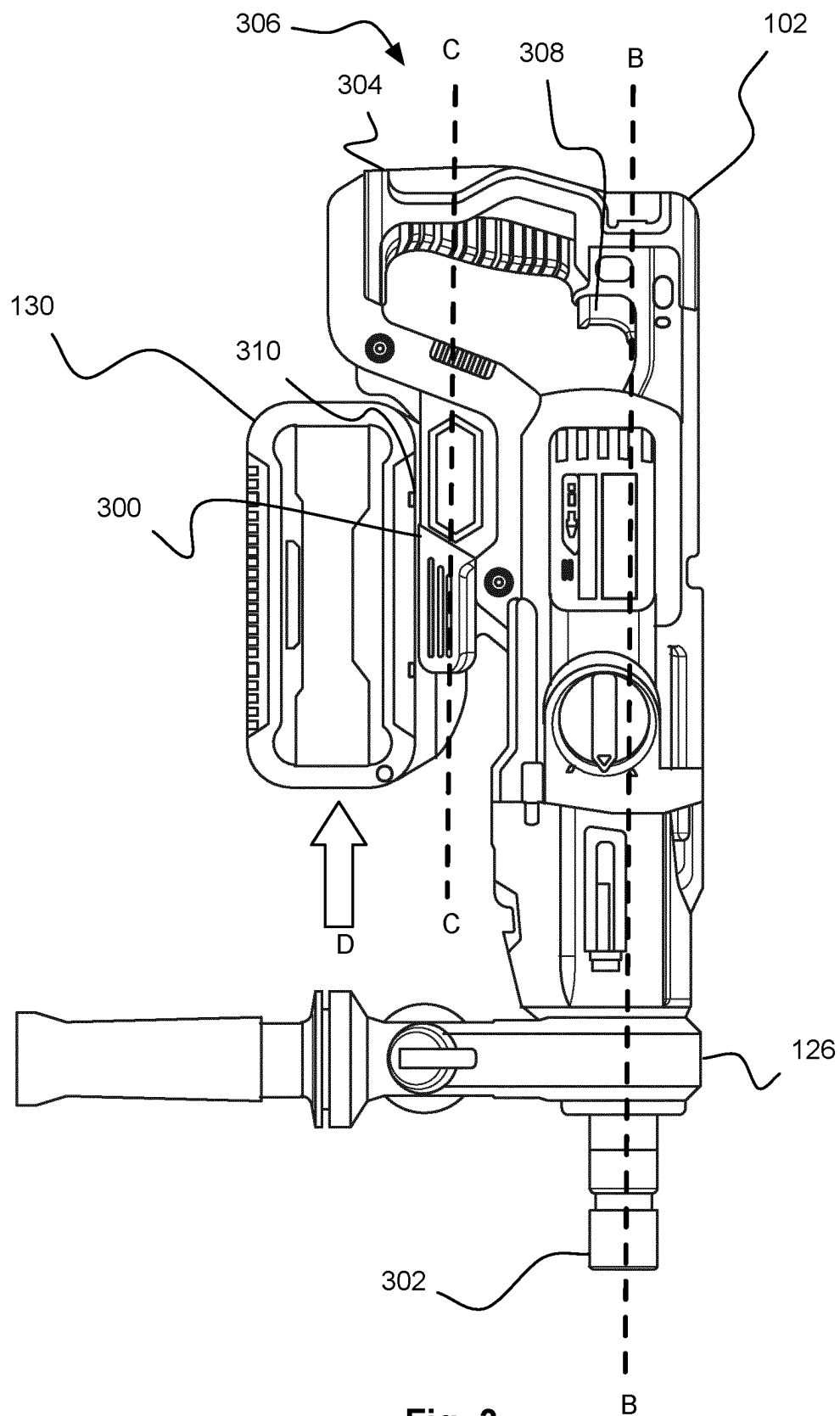


Fig. 3

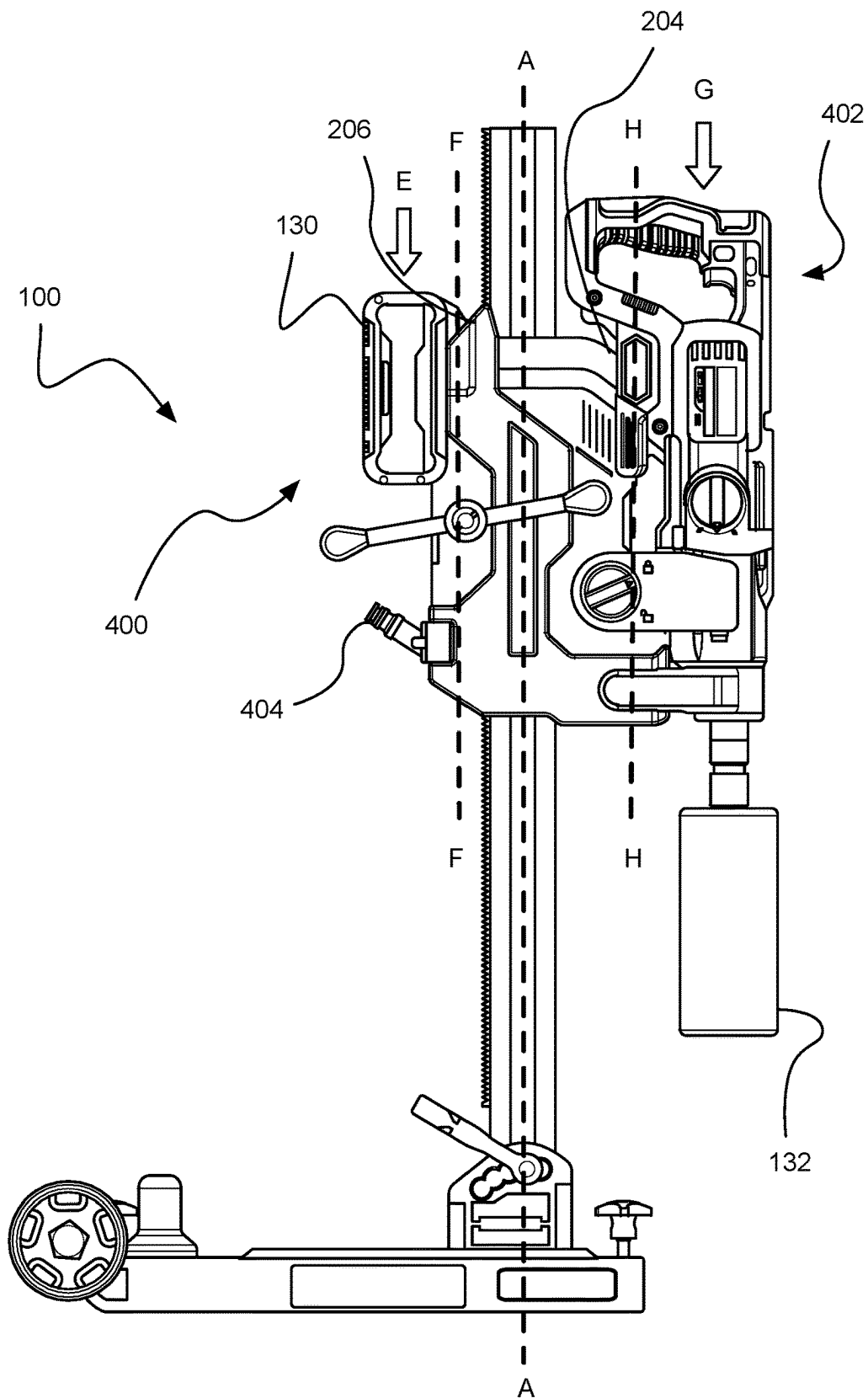


Fig. 4

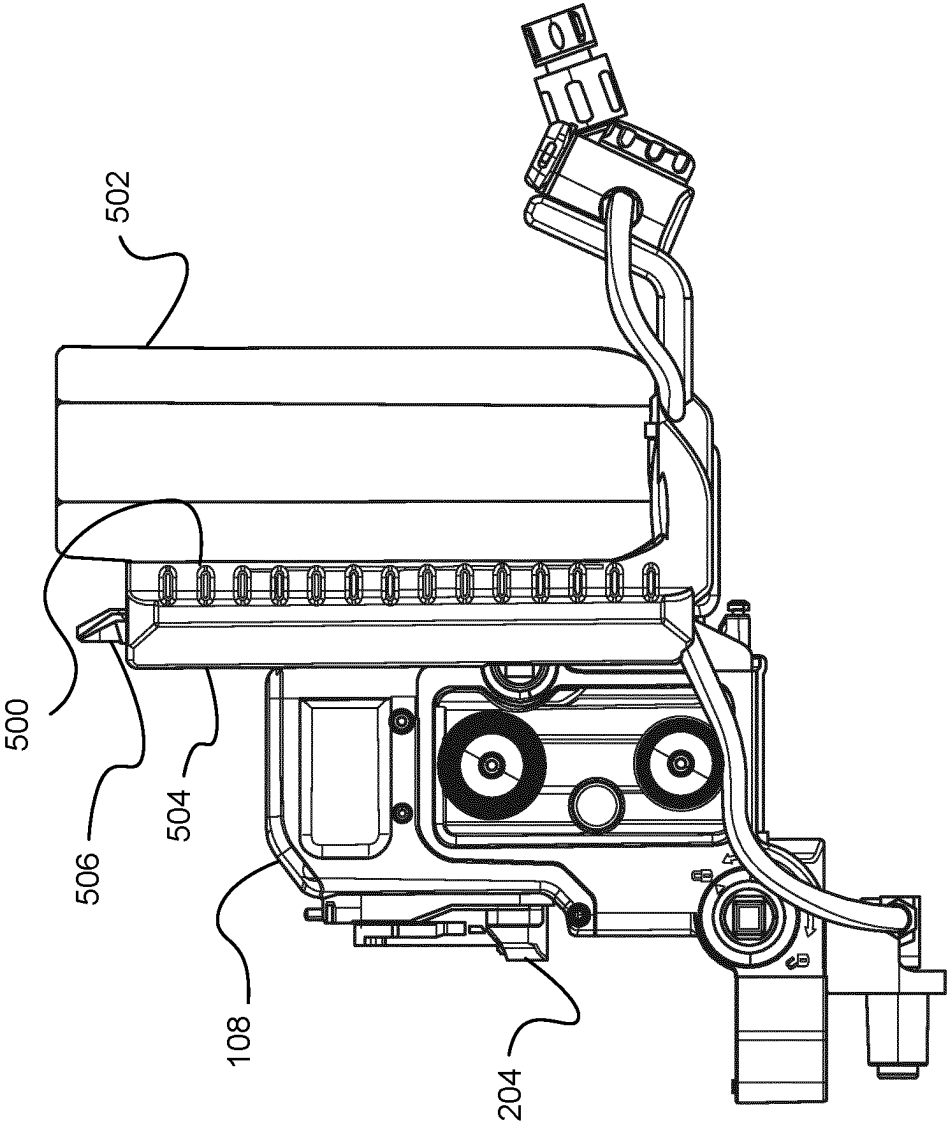


Fig. 5

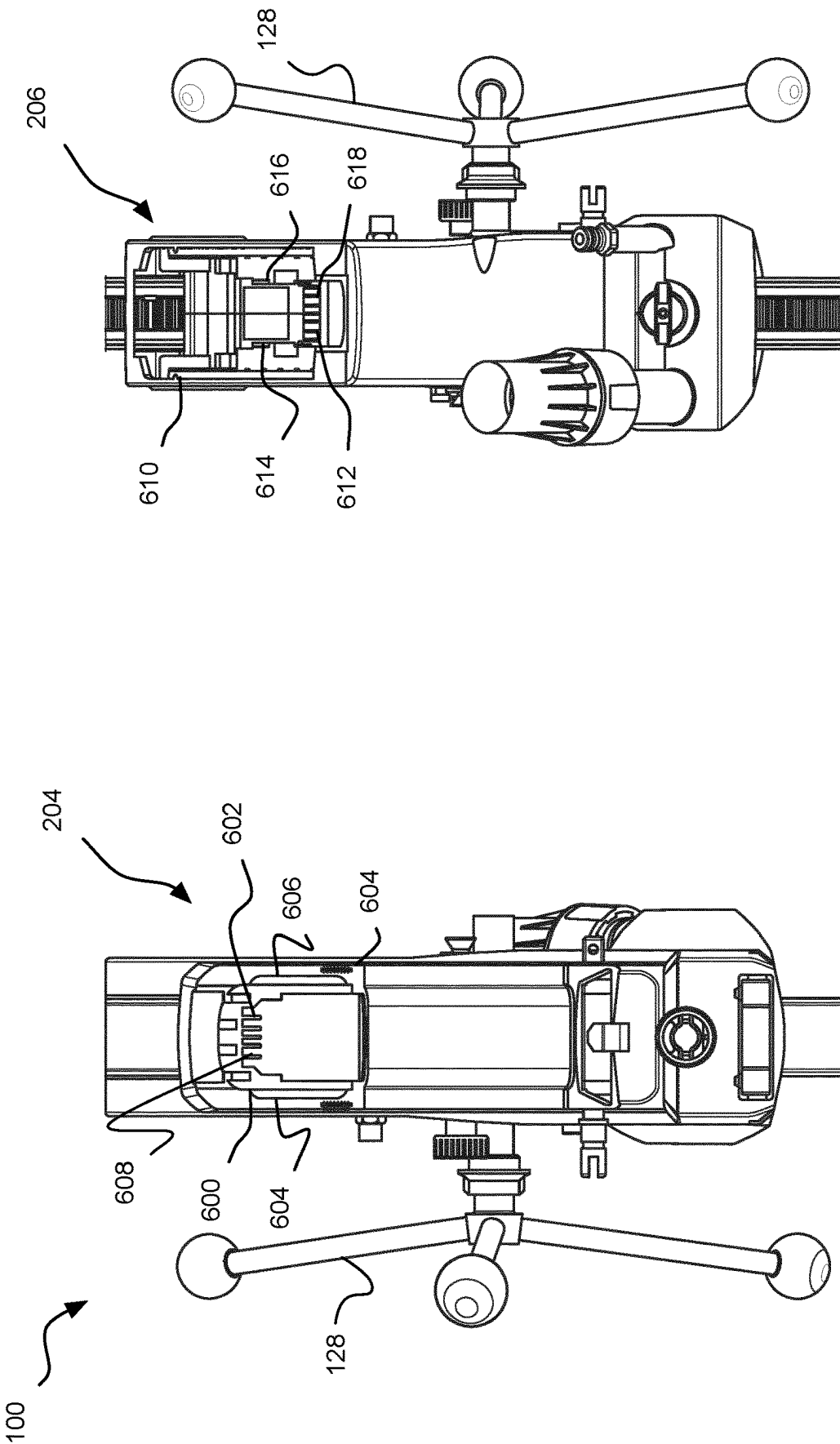


Fig. 6b

Fig. 6a

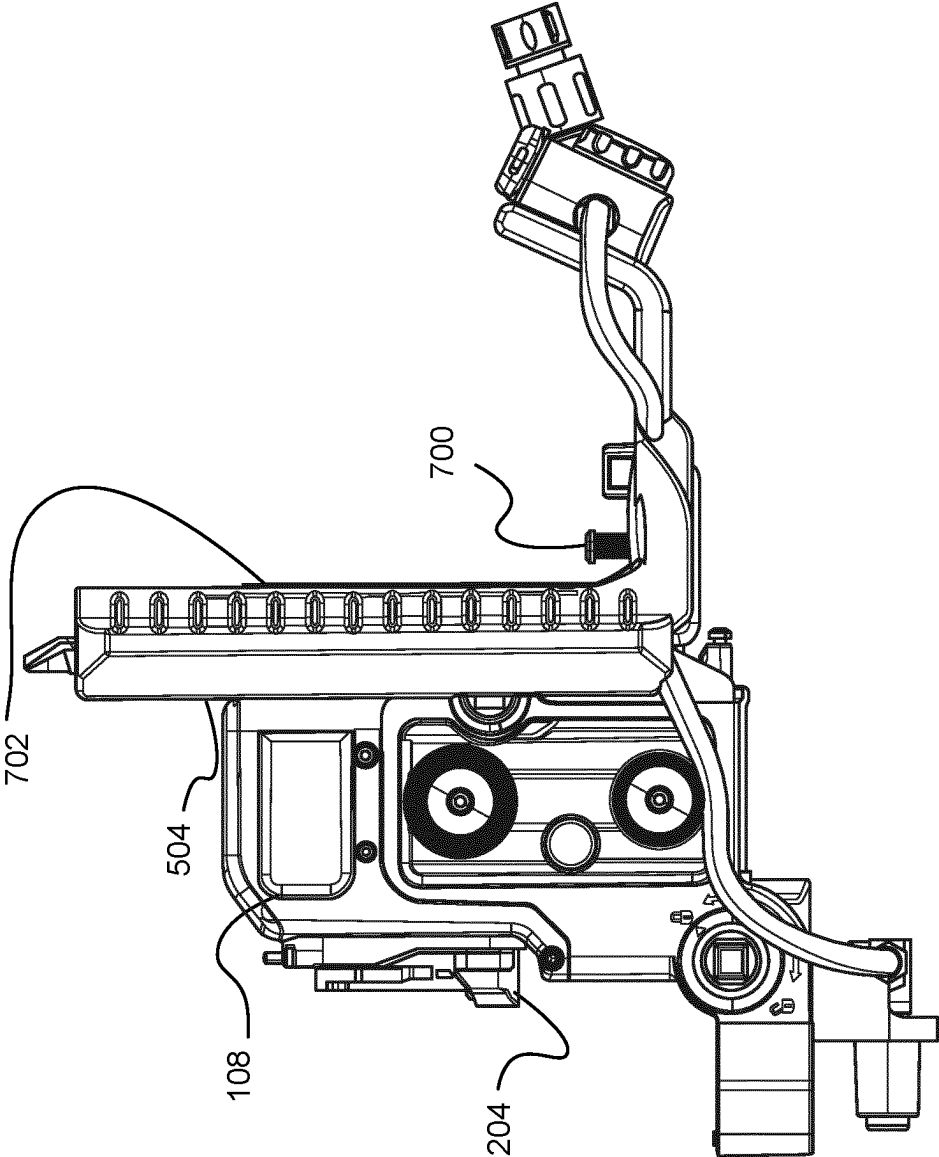


Fig. 7

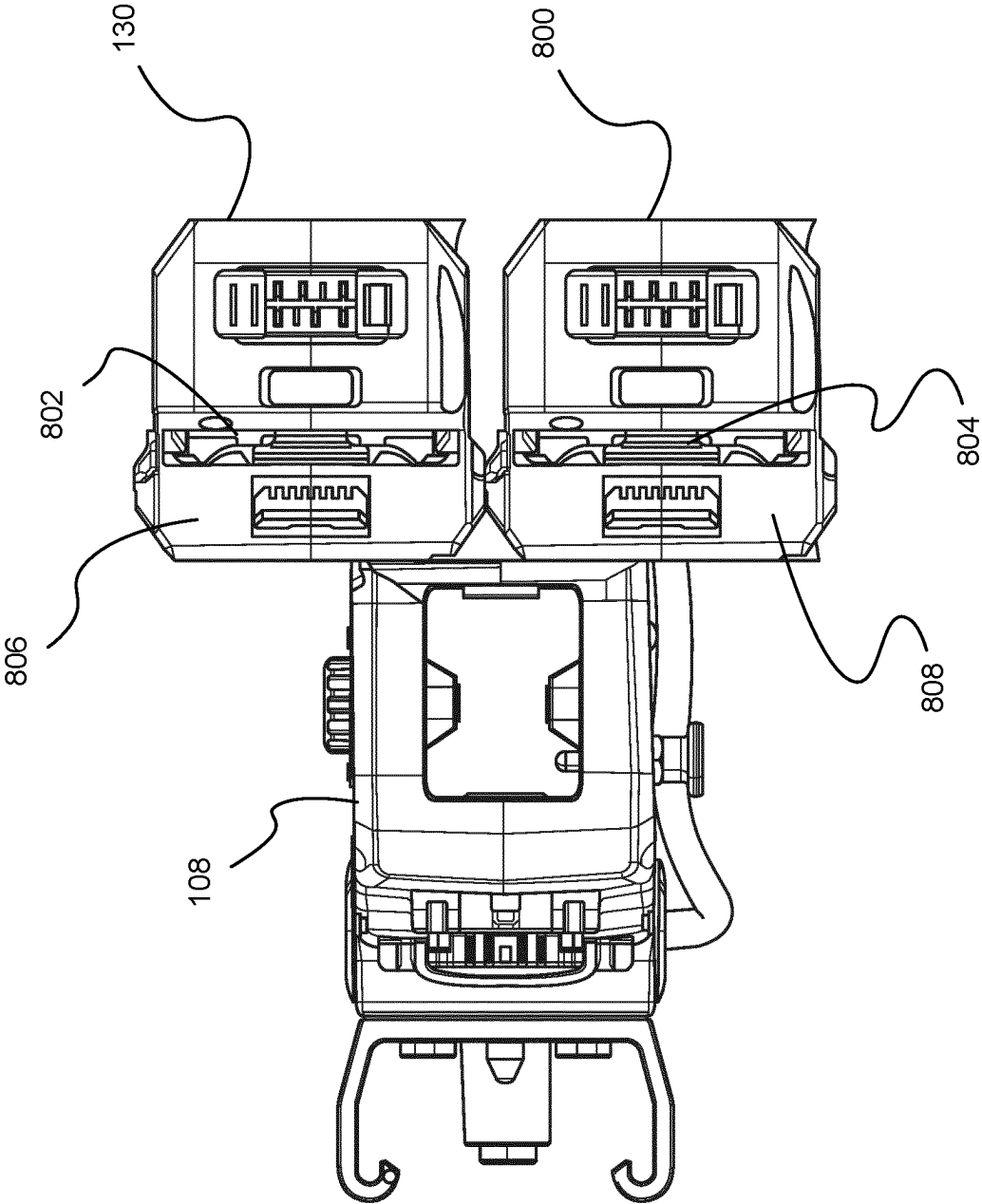


Fig. 8

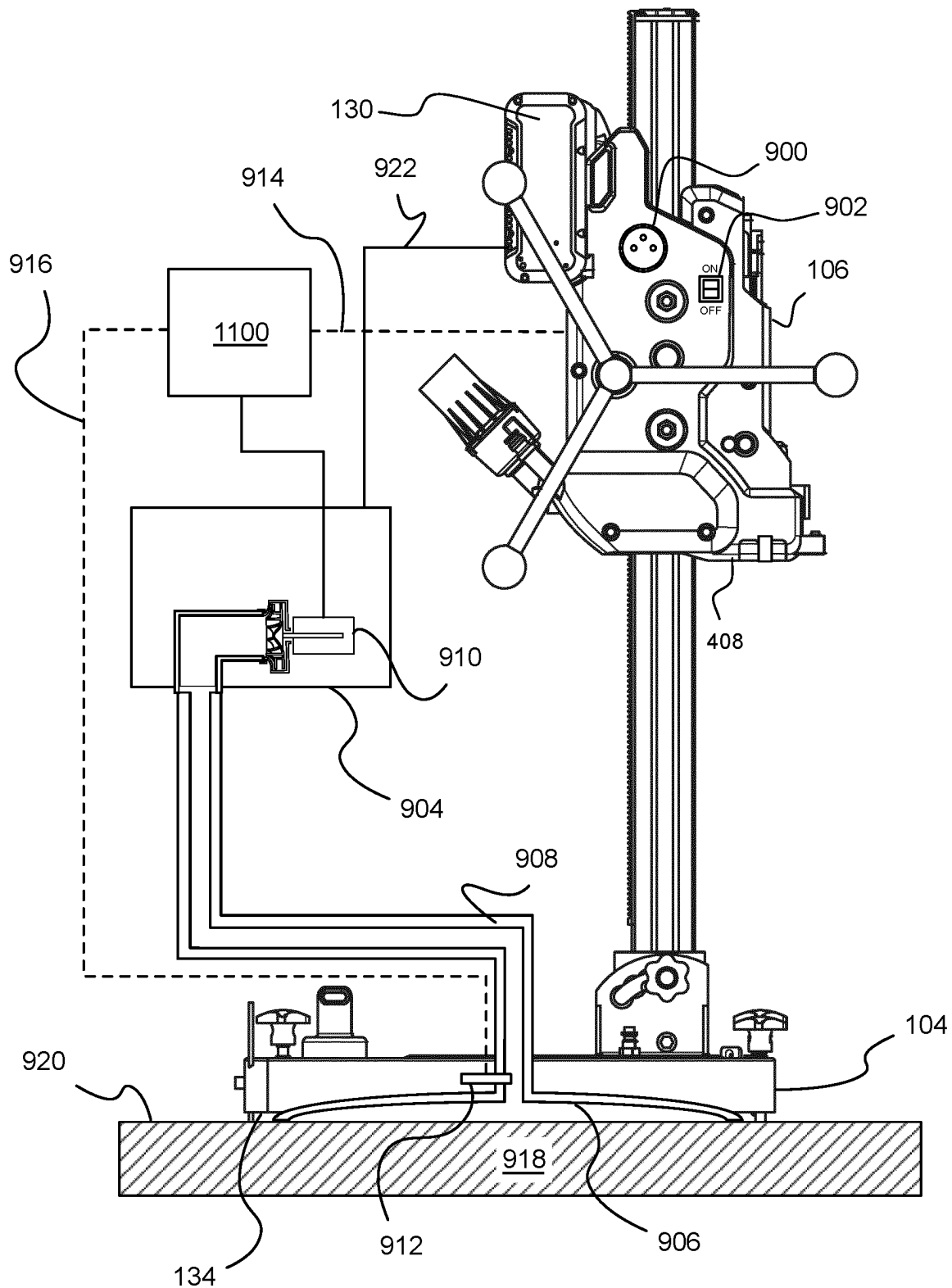


Fig. 9

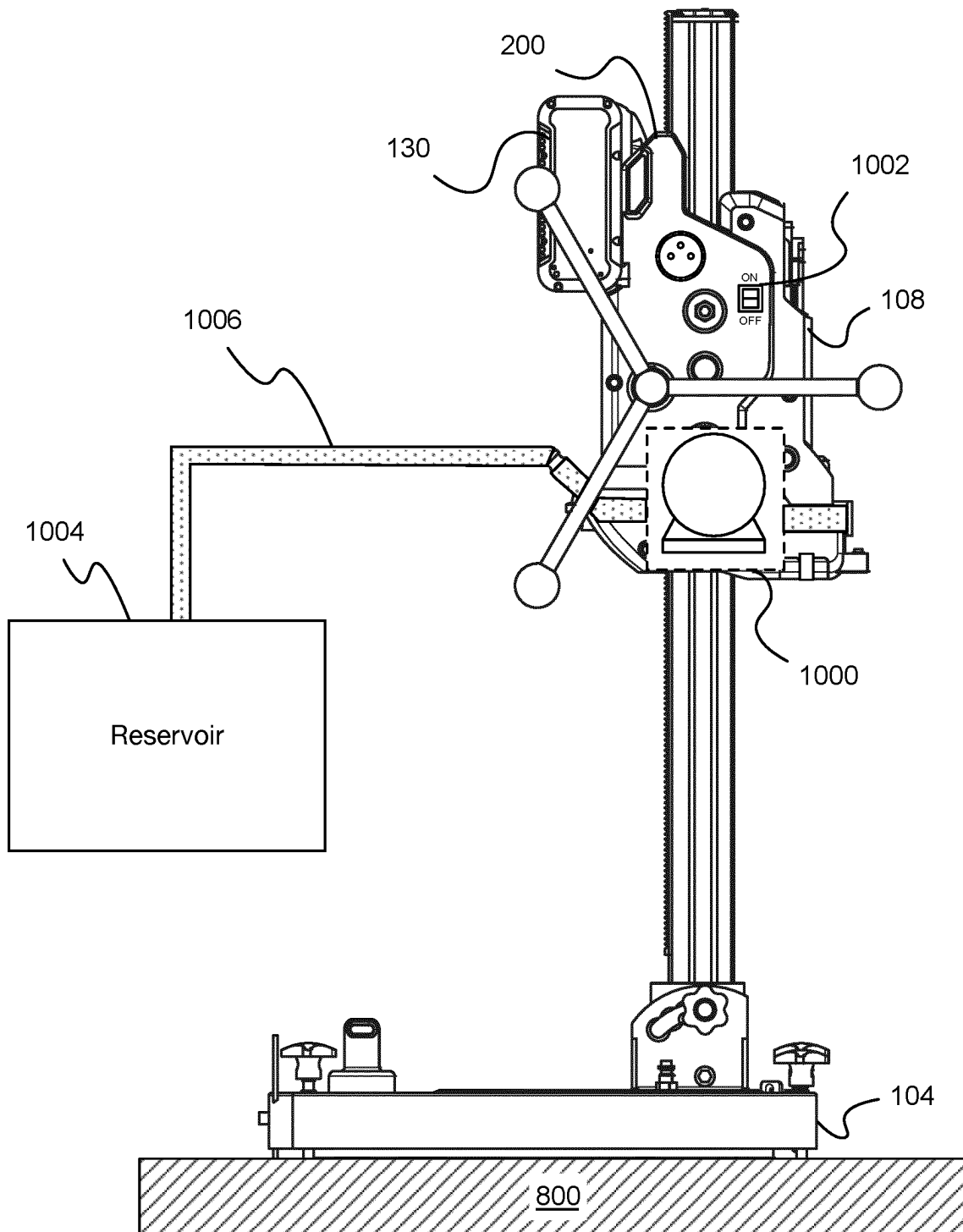


Fig. 10

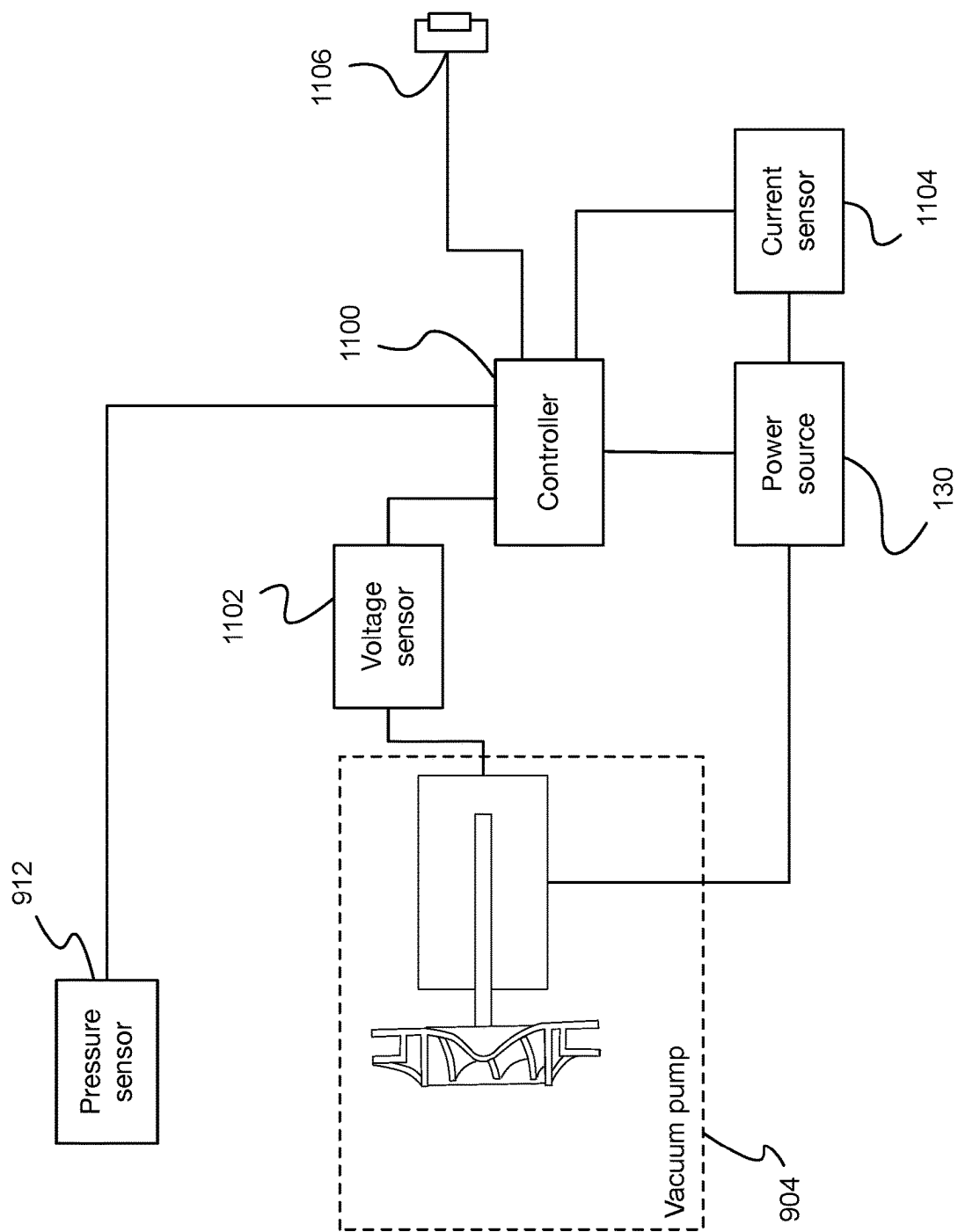


Fig. 11

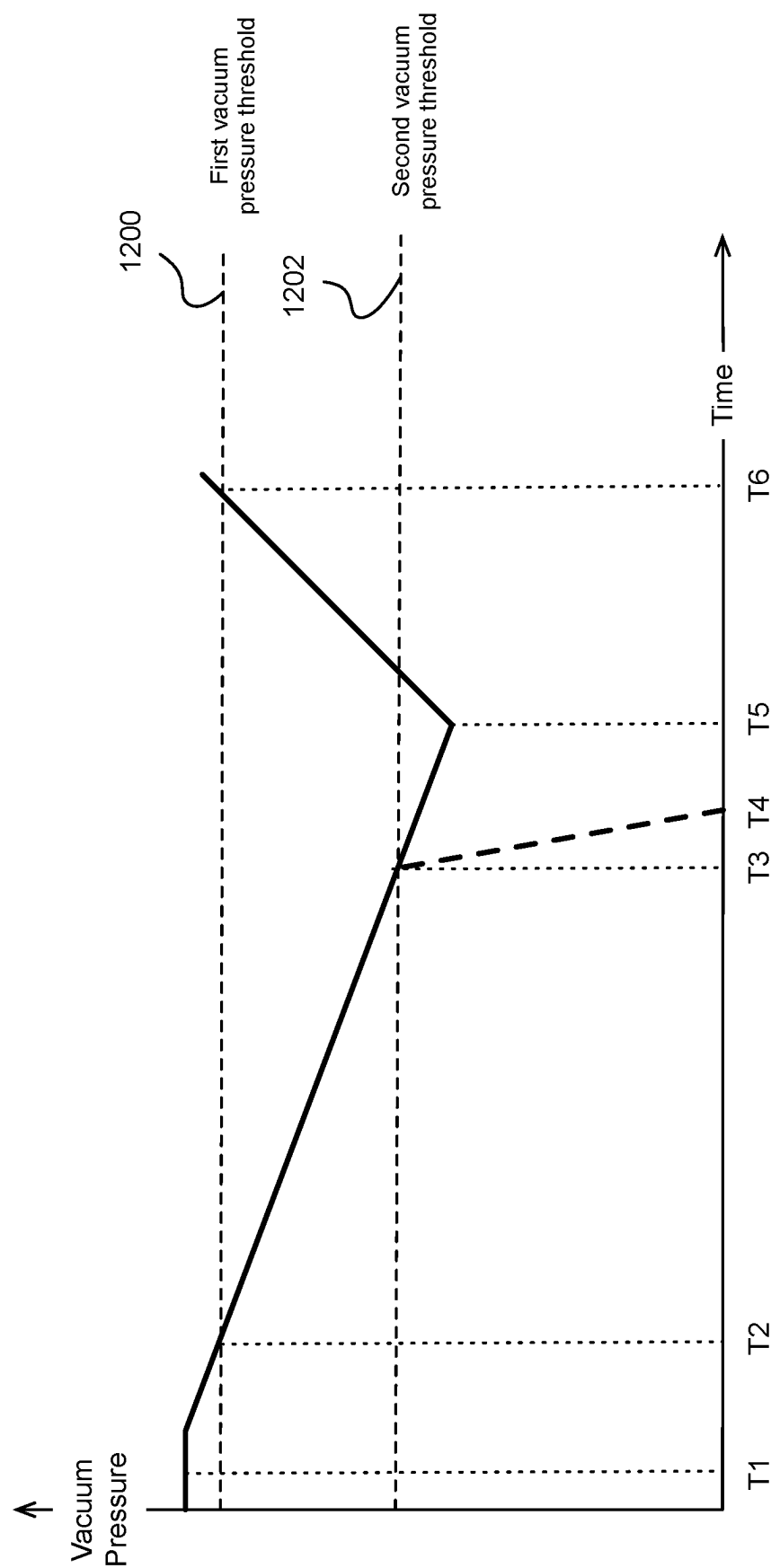


Fig. 12

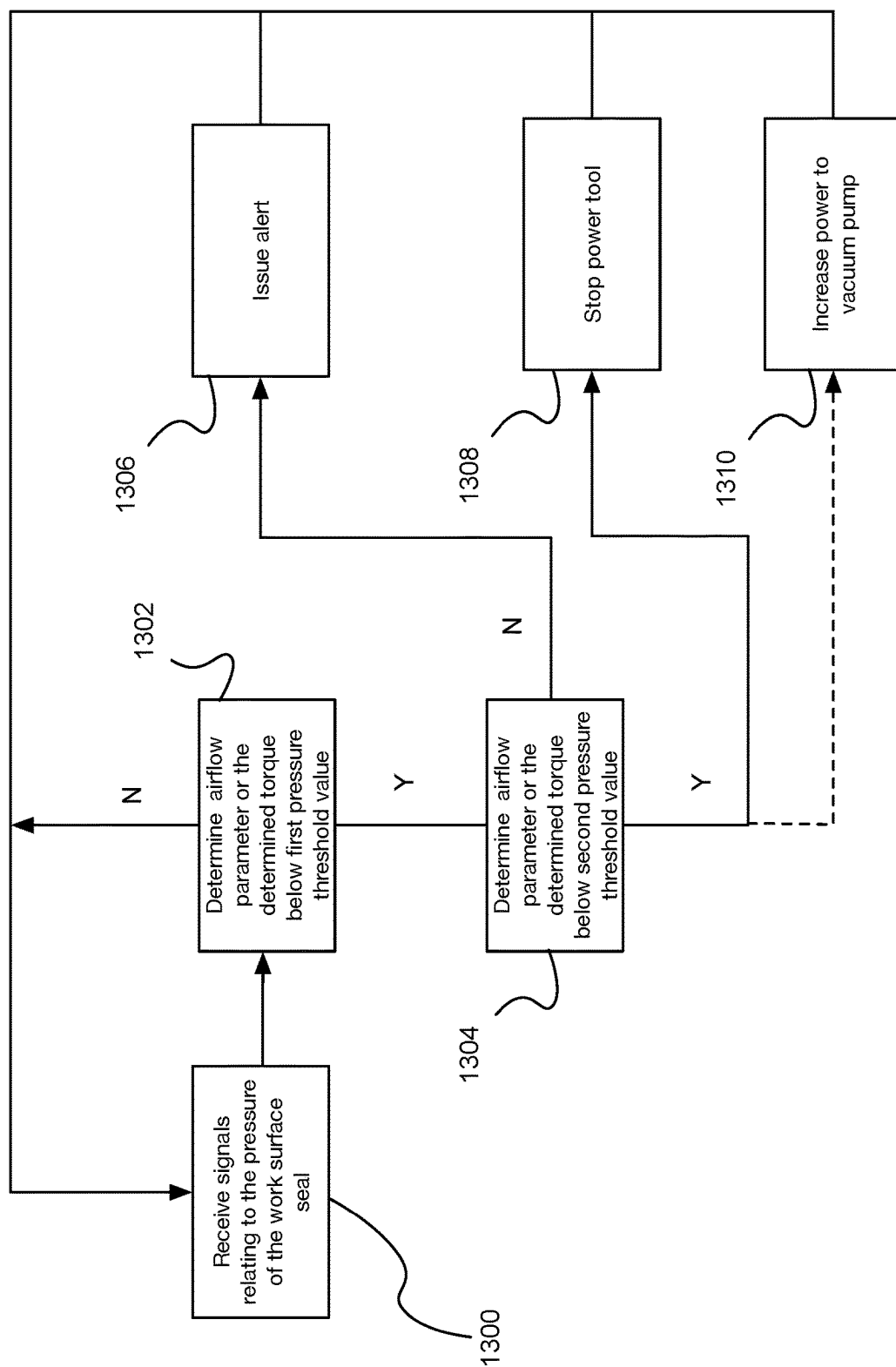


Fig. 13

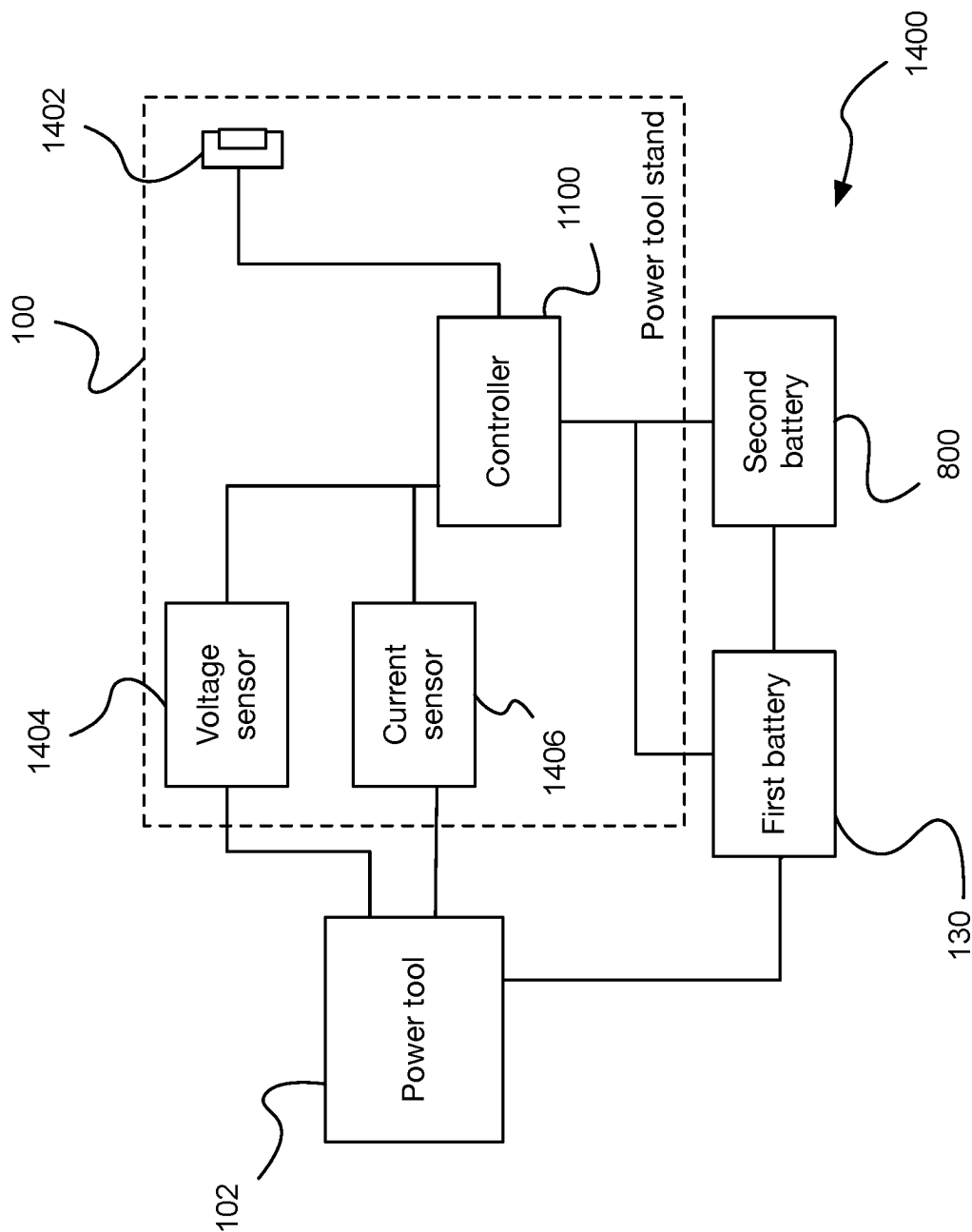


Fig. 14

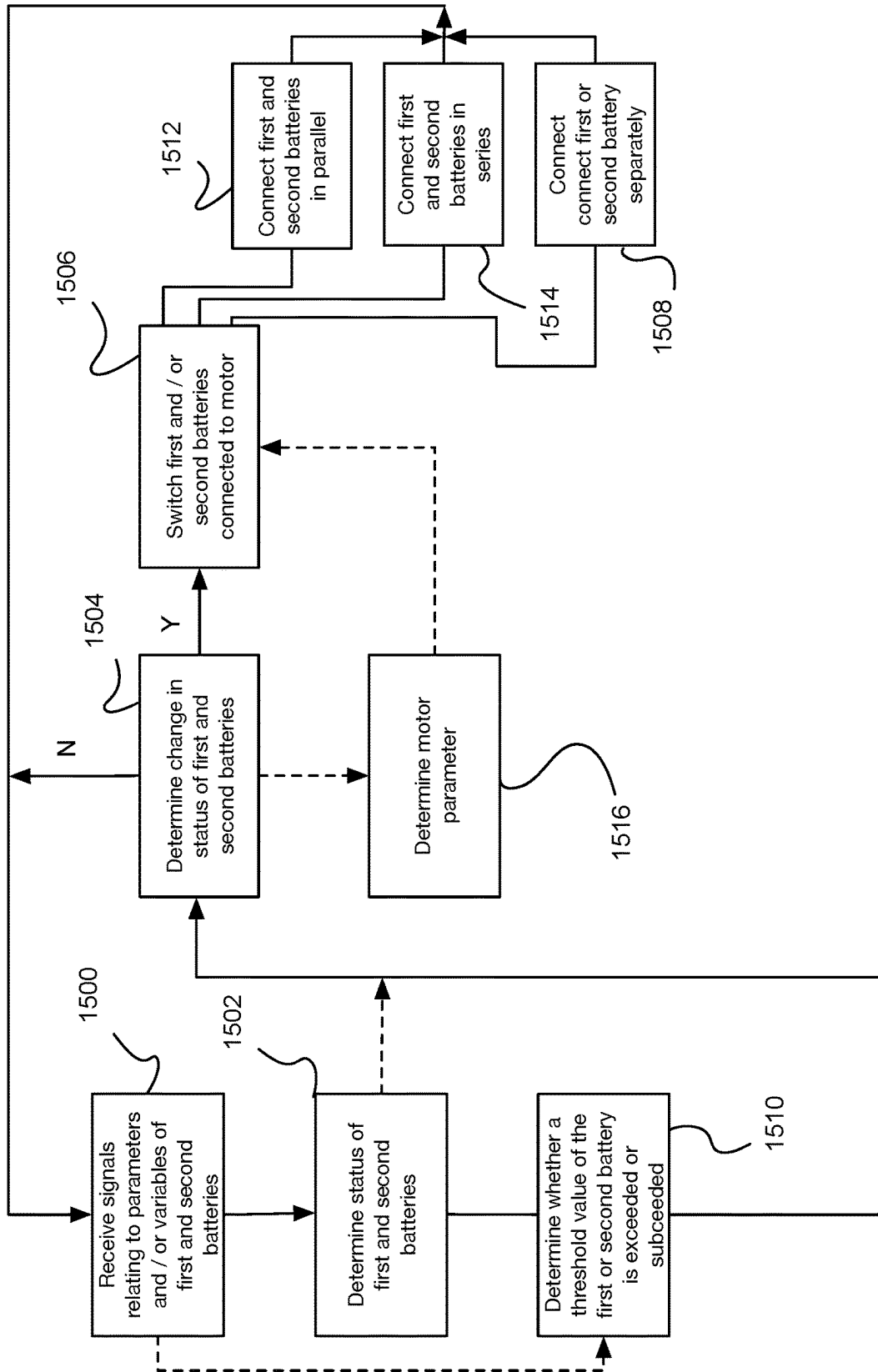


Figure 15

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POWER TOOL STAND

RELATED APPLICATION

This application is a national stage of PCT International Application No. PCT/EP2022/074710 filed Sep. 6, 2022, which claims priority to GB Patent Application No. 2112788.1 filed Sep. 8, 2021, the contents of both of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

The present disclosure relates to a power tool stand. In particular the present disclosure relates to drill power tool stand.

BACKGROUND

Some power tools are mountable on a tool stand. For example, certain drills can be mountable on a drill stand. This can improve the stability of the power tool during use and ensure that the power tool alignment with respect to the workpiece remains fixed during use. Furthermore, the power tool stand can be mountable onto the workpiece and this can further improve precision during operation.

One such power tool stand is shown in US 2019/0321896 which shows a drill stand on which a battery powered drill can be mounted. A problem with this is that the drill power tool is very large and heavy and this makes mounting and using the drill power tool on the drill power tool stand cumbersome. For example, the base of the drill power tool stand may need to be weighted down during use to stop it toppling over when used in a vertical orientation.

Once the drill described in US 2019/0321896 is mounted on the drill stand, then suitable dust management and lubrication management hoses may need to be attached to the drill. These hoses can interfere with the operation of the drill when installed on the drill stand. This may make setup, use and removal of the drill on the drill stand awkward and time consuming.

The drill stand may be mounted on a workpiece surface. It is known to use a vacuum pump with the drill stand to provide a suction force against the workpiece surface as shown US2016/151934. A problem with this is that a control unit detects signals from an accelerometer to detect vibrations in the drill stand to infer a poor seal against the workpiece surface. However, if the drill experiences vibrations due to other issues, the control unit can incorrectly control the vacuum pump.

SUMMARY

In a first aspect of the disclosure, there is provided a power tool stand for a removeable power tool having a first battery interface and a removeable battery having a first tool interface, the power tool stand comprising: a base; a projecting frame connected to the base; and a power tool carriage moveably mounted on the projecting frame, the power tool carriage comprising a second battery interface and a second tool interface; wherein the second battery interface is compatible with the first tool interface and the second battery interface is arranged to electrically and mechanically connect together with first tool interface; and the second tool interface is compatible with the first battery interface and the second tool interface is arranged to elec-

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trically and mechanically connect together with first battery interface; and the first battery interface and the first tool interface are incompatible.

Optionally, the power tool carriage comprises a battery adapter comprising the second battery interface.

Optionally, the battery adapter is mountable to the power tool carriage.

Optionally, the battery adapter comprises a carriage interface and the power tool carriage comprises an adaptor interface and adaptor interface is compatible with the carriage interface and the adaptor interface is arranged to electrically and mechanically connect together with carriage interface.

Optionally, the carriage interface is the same as the second tool interface and the adaptor interface is the same as the first battery interface.

Optionally, the battery adaptor is removably mountable to the power tool carriage.

Optionally, the second battery interface is configured to mechanically and electrically connect to a battery having a voltage of at least 60V.

Optionally, the power tool stand comprises a rack and pinion mechanism arranged to move the power tool carriage relative to the projecting frame.

Optionally, the power tool carriage comprises a handle coupled to the rack and pinion mechanism.

Optionally, the second battery interface is mounted on a first side of the power tool carriage and second tool interface is mounted on a second side of the power tool carriage.

Optionally, the second tool interface is arranged project over the base.

Optionally, the power tool stand comprises a plurality of the second battery interfaces arranged to electrically connect with a plurality of removeable batteries.

Optionally, the power tool carriage comprises a latch recess for engaging with a latch mechanism mounted on the removeable battery.

Optionally, the power tool carriage comprises an air duct connectable to a dust extractor.

Optionally, the power tool carriage comprises a water duct connectable to a water supply.

Optionally, the second battery interface is electrically connected to the second tool interface within the housing of the power tool carriage.

In a second aspect of the disclosure there is provided a power tool stand system comprising: a power tool stand according to the first aspect; a removeable power tool having a first battery interface and the first battery interface is connectable to the second tool interface of the power tool stand; and a first removable battery having a first tool interface and the first tool interface is connectable to the second battery interface of the power tool stand.

Optionally, the power tool stand system comprises a second battery having a third tool interface compatible with the first battery interface.

In a third aspect of the disclosure there is provided a power tool stand comprising: a base; a projecting frame connected to the base; and a power tool carriage moveably mounted on the projecting frame; wherein the power tool carriage comprises a tool interface arranged to electrically connect with a removeable power tool and a first battery interface and a second battery interface, the first battery interface and the second battery interface being in electrical connection with the tool interface; wherein first battery interface and the second battery interface are respectively arranged to electrically connect with a first removeable battery and a second removeable battery.

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Optionally, the first battery interface and the second battery interface are configured to connect the first removable battery and the second removable battery in series.

Optionally, the first battery interface and the second battery interface are configured to connect the first removable battery and the second removable battery in parallel.

Optionally, the power tool stand comprises a controller configured to selectively connect the first removable battery and the second removable battery to the power tool when the removable power tool, the first removable battery, and the second removable battery are mounted on the power tool carriage.

Optionally, the controller is configured to connect the first removable battery and/or the second removable battery to the removable power tool in dependence of a received signal.

Optionally, the received signal is a manually actuated signal indicating a user battery selection.

Optionally, the received signal comprises a voltage indication of the first removable battery and/or the second removable battery and the controller is configured to connect the removable power tool to the first removable battery and/or the second removable battery in dependence of the voltage indication of the first removable battery and/or the second removable battery.

Optionally, the controller is configured to connect the first removable battery and the second removable battery to the removable power tool in series, in parallel or separately.

Optionally, the controller is configured to compare the voltage indication of the first removable battery and the second removable battery and connect the first removable battery or the second removable battery in dependence of the voltage indication of the first removable battery or the second removable battery being closest to a required voltage of the removable power tool.

Optionally, wherein the required voltage of the removable power tool is in dependence of the motor speed, a motor voltage rating, or a motor parameter.

Optionally, the received signal comprises an indication that a status of either the first removable battery or the second removable battery has exceeded a threshold and the controller is configured to connect the motor to the other of the at least one first removable battery or the at least one second removable battery which has not exceeded the threshold.

Optionally, the threshold is a battery operating temperature threshold, a battery capacity threshold, a battery voltage threshold, or a battery current threshold.

Optionally, the received signal is signal indicating an error status of either the first removable battery or the second removable battery is malfunctioning or disconnected and the controller is configured to connect the removable power tool to the other of the first removable battery or the second removable battery.

Optionally, the first battery interface and the second battery interface are mounted on a first side of the power tool carriage and tool interface is mounted on a second side of the power tool carriage.

Optionally, the tool interface is arranged to project over the base.

In a fourth aspect of the disclosure, there is provided a power tool stand comprising: a base; a projecting frame connected to the base; a power tool carriage moveably mounted on the projecting frame; a tool interface mounted on the power tool carriage arranged to couple with a removable power tool; an electrical interface mounted on the power tool carriage arranged to electrically couple with

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a removable battery; wherein the power tool stand comprises at least one duct configured to convey a fluid; and at least one pump in fluid communication with the fluid duct wherein the at least one pump is in electrical connection with the electrical interface.

Optionally, the at least one duct is connected to a fluid connector configured to engage with a power tool fluid duct when the removable power tool is mounted to the power tool carriage.

Optionally, the at least one duct is in fluid communication with a water supply.

Optionally, the water supply is a non-pressurised water supply.

Optionally, the water supply is a water reservoir.

Optionally, power tool carriage comprises a switch in electrical connection with the at least one pump configured to selectively control actuation of the at least one pump.

Optionally, the switch is configured to selectively control actuation of the at least one pump when the removable power tool and the removable battery are mounted on the power tool carriage.

Optionally, the at least one pump is mounted in the power tool carriage.

Optionally, the at least one duct is mounted on the power tool carriage.

Optionally, the power tool stand comprises a controller configured to selectively control actuation of the at least one pump.

Optionally, the tool interface comprises a power tool electrical interface mounted on the power tool carriage arranged to electrically couple with the removable power tool.

Optionally, the power tool electrical interface is in electrical connection with the electrical interface.

In a fifth aspect of the disclosure, there is provided a power tool stand comprising: a base mountable on a workpiece surface; a frame connected to the base; a power tool carriage mounted on the frame; a tool interface mounted on the power tool carriage arranged to secure a removable power tool; a work surface seal mounted on the base and configured to engage the workpiece surface and in fluid communication with a vacuum source and the work surface seal is configured to generate a vacuum seal against the workpiece surface when the vacuum source is actuated; a pressure sensor in fluid communication with the work surface seal and configured to generate a pressure signal; and a controller configured to determine when the vacuum pressure in work surface seal is decreasing in dependence of the received pressure signal from the pressure sensor and issue one or more control signals based on a detected decreased vacuum pressure in the work surface seal; wherein the power tool carriage comprises a battery interface arranged to secure a removable battery and the battery interface is electrically connected to the tool interface.

Optionally, the controller is configured to send a control signal to the removable power tool.

Optionally, the controller is configured to issue a first control signal in dependence of determining that the vacuum pressure in the work surface seal falls below a first vacuum pressure threshold.

Optionally, the first control signal is configured to issue an alert to the user when the vacuum pressure in the work surface seal falls below a first vacuum pressure threshold.

Optionally, the controller is configured to issue a second control signal in dependence of determining that the vacuum pressure of the work surface seal falls below a second

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vacuum pressure threshold, wherein the second vacuum pressure threshold is lower than the first vacuum pressure threshold.

Optionally, the second control signal is configured deactivate the removable power tool.

Optionally, the power tool carriage comprises a switch in electrical connection with the tool interface and the battery interface configured to selectively control actuation of the removable power tool and the controller is configured to actuate the switch.

Optionally, the vacuum source is a vacuum pump.

Optionally, the vacuum pump is mounted in the power tool carriage or on the base.

Optionally, the vacuum pump is electrically connected to the removable battery when the removable battery is mounted to the battery interface.

Optionally, the controller is configured to issue a control signal to the vacuum pump to increase the vacuum pressure in the work surface seal.

Optionally, the pressure sensor is wirelessly connected or in wired connection with the controller.

Optionally, the work surface seal is configured mount to a vertical surface.

In a sixth aspect of the disclosure, there is provided a power tool stand comprising: a base mountable on a workpiece surface; a frame connected to the base; a power tool carriage mounted on the frame; a tool interface mounted on the power tool carriage arranged to secure a removable power tool; a work surface seal mounted on the base and configured to engage the workpiece surface and in fluid communication with a vacuum source and the work surface seal is configured to generate a vacuum seal against the workpiece surface when the vacuum source is actuated; wherein the vacuum source is a vacuum pump mounted in the power tool stand.

Optionally, the vacuum pump is mounted in the power tool carriage.

Optionally, the vacuum pump is mounted in the base.

BRIEF DESCRIPTION OF THE DRAWINGS

Various other aspects and further examples are also described in the following detailed description and in the attached claims with reference to the accompanying drawings, in which:

FIG. 1 shows a perspective view of a power tool stand according to an example;

FIG. 2 shows a perspective view of part of a power tool stand according to an example;

FIG. 3 shows a side view of an exemplary power tool mountable on a power tool stand according to an example;

FIG. 4 shows a side view of a power tool stand according to an example;

FIG. 5 shows a side view of part of a power tool stand according to an example;

FIGS. 6a and 6b respectively show a partial front and rear view of a power tool stand according to an example;

FIG. 7 shows a side view of part of a power tool stand according to an example;

FIG. 8 shows a plan view of part of a power tool stand according to an example;

FIG. 9 shows a side view of a power tool stand according to an example;

FIG. 10 shows a side view of a power tool stand according to an example;

FIG. 11 shows a schematic view of a power tool stand according to an example;

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FIG. 12 shows a graph of pressure of a workpiece surface seal over time according to an example;

FIG. 13 shows a flow diagram of operation of a power tool stand according to an example;

FIG. 14 shows a schematic view of a power tool stand according to an example; and

FIG. 15 shows a flow diagram of operation of a power tool stand according to an example.

DETAILED DESCRIPTION

FIG. 1 shows a perspective view of a power tool stand 100 according to an example. The power tool stand 100 is arranged to receive a removable power tool 102. The removable power tool 102 can be mounted and secured in the power tool stand 100 as required by the user. The term “removable” refers to the function of the removable power tool 102 being selectively removable from the power tool stand 100. In other words, the removable power tool 102 is not permanently fixed to the power tool stand 100.

The power tool stand 100 comprises a base 104 and a projecting frame 106 fixed to the base 104. A power tool carriage 108 is moveably mounted on the projecting frame 106 and moveable along a longitudinal axis A-A of the projecting frame 106. By moving the power tool carriage 108 along the projecting frame 106, the distance between the power tool carriage 108 and the base 104 can be varied. The projecting frame 106 projects through a through hole 202 (best shown in FIG. 2) in the power tool carriage 108. FIG. 2 shows a perspective view of the power tool carriage 108 of the power tool stand 100.

The power tool carriage 108 is optionally moveably coupled to projecting frame 106 via a carriage mechanism (not shown). In some less preferred examples (not shown in the Figs), the power tool carriage 108 is fixed with respect to the projecting frame 106 and does not move with respect to the power tool carriage 108.

The projecting frame 106 as shown in FIG. 1 is substantially upright. In some examples, the power tool stand 100 can be used in the orientation as shown in FIG. 1 and in this case, the projecting frame 106 will remain substantially upright positioned on a workpiece surface 920 of a workpiece 918. The workpiece surface 920 is best shown in FIG. 8. FIG. 8 shows a plan view of the power tool carriage 108 of the power tool stand 100.

However, in some other examples, the power tool stand 100 can be used in a substantially horizontal orientation. In this case, the projecting frame 106 will be substantially horizontal. Indeed, the power tool stand 100 may be used in any orientation as required and the position of the projecting frame 106 will be adjusted accordingly.

The projecting frame 106 as shown in FIG. 1 is a single post projecting up from the base 104. However, in other examples the projecting frame 106 can optionally comprise a plurality of posts (not shown) projecting up from the base 104 and the power tool carriage 108 may be mounted on the plurality of posts. The projecting frame 106 as shown in FIG. 1 is mounted perpendicular to the plane of the base 104 at a coupling 110. The plane of the base 104 can be substantially parallel with the plane of a workpiece 918 (best shown in FIG. 9). FIG. 9 shows a side view of the power tool stand 100 with the removable power tool 102 mounted thereon.

In some examples, the coupling 110 optionally comprises a pivot connection and the projecting frame 106 is pivotable with respect to the base 104. In this way, the projecting frame 106 can be angled with respect to a normal axis (not

shown) of the plane of a workpiece **918**. The normal axis of the plane of the workpiece **918** and the base **104** is parallel with the longitudinal axis A-A as shown in FIG. 1.

The coupling **110** comprises a screw fastener **112** threaded through arcuate slots **114** in a pair of projecting plates **116** fixed to the base **104**. The screw fastener **112** is screwed into the projecting frame **106** or into a reciprocal fastening nut **118** (as shown in FIG. 2) on the opposite side of the projecting frame **106**. When the screw fastener **112** and reciprocal fastening nut **118** are tightened against the projecting plates **116** and the projecting frame **106**, frictional forces keep the projecting frame **106** fixed with respect to the base **104**. Other fastening mechanisms e.g. clips, clamps, or any other suitable means can be used to fix the projecting frame **106** with respect to the base **104**.

The pair of projecting plates **116** each comprises an arcuate slot **114** and form a yoke for receiving a portion of the projecting frame **106** between the projecting plates **116**. The yoke as shown in FIG. 1 is optional and other pivot connections can be provided. For example a single projecting plate **116** can be used to couple the base **104** to the projecting frame **106**. In some examples, the arcuate slot **114** optionally comprises a plurality of indexing positions (not shown) for positioning the reciprocal fastening nut **118** and the screw fastener **112** at predetermined positions along the arcuate slot **114**.

Optionally, the projecting plates **116** comprise an inclination scale (not shown) to indicate the angle of the projecting frame **106** with respect to the plane of the base **104**. In some examples the projecting frame **106** is arranged to pivot between 0 to 60 degrees from a normal axis (not shown) perpendicular to the base **104**. In other examples, the projecting frame **106** can pivot any angle with respect to a normal axis perpendicular to the base **104**.

In some examples, the projecting frame **106** is alternatively permanently fixed with respect to the base **104**. In this case the projecting frame **106** cannot be adjusted with respect to the base **104**. For example, the projecting frame **106** is welded to the base **104** in the upright position as shown in FIG. 1.

In some examples, the projecting frame **106** can be pivotable with respect to base **104** such that the longitudinal axis A-A of the projecting frame **106** can be positioned parallel with the plane of the base **104**. This can be useful for transporting the power tool stand **100** to the work site.

In some examples, the projecting frame **106** comprises a reinforced cross-sectional shape for increasing rigidity of the projecting frame **106**. The projecting frame **106** can optionally comprise an I-beam cross-sectional shape in some examples.

In some examples, the base **104** comprise one or more fixing bolts **120** for anchoring the base **104** of the power tool stand **100** to the workpiece **918**. FIG. 1 shows the base **104** comprising four fixing bolts **120** located at each corner of the base **104**. In some examples, there can be additional or fewer fixing bolts **120** for anchoring the base **104** to the workpiece **918** as required.

The fixing bolts **120** allow the power tool stand **100** to be securely fastened to a workpiece **918** surface in a horizontal plane, a vertical plane, or an inclined surface. For example a user may using the fixing bolts **120** to anchor the power tool stand **100** to a floor or a wall before operation.

When the user anchors the base **104** to a vertical surface such as a wall (not shown), the user may disassemble the power tool stand **100** and remove the projecting frame **106**

from the base **104**. Once the user has anchored the base **104** to the wall, the user may then reattach the projecting frame **106** back on the base **104**.

The base **104** optionally comprises wheels **122** for transporting the power tool stand **100** to the worksite. The wheels **122** are mounted on the base **104** such that the wheels **122** do not project beyond an engaging surface **134** of the base **104**. The engaging surface **134** is arranged to abut the workpiece surface **920**. Accordingly, the wheels **122** do not touch the workpiece surface **920** when the engaging surface **134** abuts the workpiece surface **920**. In some examples, the wheels **122** are optionally removeable. This means the user can remove the wheels **122** when placing the power tool stand **100** in tight spaces on a wall or a floor.

The base **104** can optionally further comprise one or more stand handles (not shown) for also assisting transportation and handling. The user can optionally grasp the stand handle mounted on the base **104** and the projecting frame **106** in order to move the power tool stand **100** into position.

The power tool carriage **108** and power tool stand **100** will be discussed in more detail with respect to FIGS. 1 and 2. As shown in FIG. 2, the power tool carriage **108** is arranged to receive a removeable power tool **102** and a removeable battery **130**. The projecting frame **106** allows a power tool carriage **108** to be precisely moved along the longitudinal axis A-A and then fixed with respect to the projecting frame **106**.

The power tool carriage **108** comprises a carriage housing **200**. The carriage housing **200** comprises a clam shell type construction having two halves which are fastened together. The halves of the carriage housing **200** are fastened together with screws but in alternative examples any suitable means for fastening the carriage housing **200** together may be used such as glue, clips, bolts and so on. For the purposes of clarity, the fastenings in the carriage housing **200** are not shown in FIG. 2. The carriage housing **200** can comprise a unitary element surrounding the internal components of the power tool carriage **108**. In other examples, the carriage housing **200** can comprise one or more housing portions (not shown) which are mounted together to form the carriage housing **200**.

The power tool carriage **108** comprises a power tool interface **204** arranged to electrically and mechanically connect the removeable power tool **102** to the power tool carriage **108**.

The power tool carriage **108** further comprises a battery interface **206** arranged to electrically and mechanically connect the removeable battery **130** to the power tool carriage **108**.

The power tool interface **204** and the battery interface **206** are electrically connected together via one or more internal wires (not shown) routed inside the carriage housing **200**. The internal wires can be for example a flexible connection ribbon arranged to electrically couple pairs of electrical contacts on the power tool interface **204** and the battery interface **206**. In other examples, power tool interface **204** and the battery interface **206** can be electrically connected by separate wires, a printed circuit board or any other suitable means for electrically connecting the power tool interface **204** and the battery interface **206**.

In this way, the removeable power tool **102** and the removeable battery **130** are respectively electrically connected to the power tool interface **204** and the battery interface **206** when mounted to the power tool carriage **108**. This means that when mounted to the power tool carriage **108**, the removeable battery **130** can supply power to the removeable power tool **102**. Accordingly, the removeable

power tool 102 can be powered by the removeable battery 130 when both mounted on the power tool carriage 108 as if the removeable battery 130 and the removeable power tool 102 are directly coupled together.

In some examples, operation of the removeable power tool 102 when mounted on the power tool stand 100 is no different from when the removeable power tool 102 is directly connected to the removeable battery 130 as shown in FIG. 3. FIG. 3 shows a side view of the removable power tool 102 with a removeable battery 130 mounted to the removeable power tool 102. The user can therefore intuitively use the removeable power tool 102 when mounted on the power tool stand 100.

Turning to FIG. 3, the removeable power tool 102 will now be discussed in more detail. FIG. 3 shows a side view of an the removeable power tool 102.

The removeable power tool 102 as shown in FIG. 3 is an example of a removeable power tool 102 that is mountable on the power tool stand 100. The removeable power tool 102 as shown in FIG. 3 in some examples is a drill. In particular, the removeable power tool 102 is a diamond core drill. In other examples, the removeable power tool 102 can be hammer drill, a router, a reciprocating saw, plunge saw, a circular saw, an impact driver, a drill, a hammer drill, a multitool, an oscillating tool, a rotary hammer, a chipping hammer, or any other similar power tool mountable on a power tool stand 100.

The removeable power tool 102 is battery powered and comprises a tool mounted battery interface 300 for engaging with the reciprocal battery mounted tool interface 310 on the removeable battery 130. The removeable power tool 102 is generally elongate and comprises a longitudinal axis B-B. The removeable power tool 102 and the removeable battery 130 are engageable along a third interface plane C-C. The third interface plane C-C is parallel with the longitudinal axis B-B of the removeable power tool 102. This means that the removeable battery 130 is mounted on the tool mounted battery interface 300 by sliding the removeable battery 130 in a direction D indicated by the arrow in FIG. 3. The direction D is away from the nose 302 of the removeable power tool 102 and towards a rear power tool handle 304 located at the rear 306 of the removable power tool 102.

In order to mount the removeable power tool 102 and the removeable battery 130 on the power tool stand 100, the removeable battery 130 is disengaged from the removeable power tool 102. This is achieved by reversing the sliding action described above in reference to FIG. 3. The separate removeable power tool 102 and the removeable battery 130 can then be individually mounted on the power tool carriage 108.

As discussed above, both the removeable power tool 102 and the removeable battery 130, are mechanically secured to the power tool carriage 108. This means that the power tool carriage 108 moves in unison with the removeable power tool 102 and the removeable battery 130. This can help the user stabilise the removeable power tool 102 during use and improves the quality of the cut in the workpiece 918.

The power tool interface 204 mounted on the power tool carriage 108 is compatible with the tool mounted battery interface 300. Likewise the battery interface 206 on the power tool carriage 108 is compatible with the battery mounted tool interface 310.

However, in some examples, the tool mounted battery interface 300 on the removeable power tool 102 is not compatible with the battery mounted tool interface 500 on the removeable battery 502 (e.g. as shown in FIG. 5).

For example, the removeable battery 502 as shown in FIG. 5 is a high capacity removeable battery 502 configured to power the removeable power tool 102 when mounted on the power tool stand 100. The high capacity removeable battery 502 may not be suitable for connecting directly to the removeable power tool 102 because the high capacity removeable battery 502 is too large and heavy. In some examples, the mechanical and electrical interfaces of the battery mounted tool interface 500 on the high capacity removeable battery 502 are different from the mechanical and electrical interfaces on the tool mounted battery interface 300 on the removeable power tool 102. That is, the removeable battery 103 as shown in FIG. 3 is different from the high capacity removeable battery 502 as shown in FIG. 5.

This means that that the high capacity removeable battery 502 as shown in FIG. 5 comprises an electrical and mechanical lock out preventing mounting of the high capacity removeable battery 502 on the removable power tool 102. This means only the correct form, shape, and specification of the removeable battery 130 and the high capacity removeable battery 502 can be mounted to the removeable power tool 102 and the power tool stand 100 respectively.

For example, the removeable battery 130 as shown in FIG. 1 is mountable directly to the removeable power tool 102, but the high capacity removeable battery 502 as shown in FIG. 5 is not mountable directly to the removable power tool 102. Instead, the high capacity removeable battery 502 as shown in FIG. 5 is mountable directly to power tool carriage 108.

In some examples, the power tool carriage 108 optionally comprises a battery adaptor 504. The battery adaptor 504 is configured to mechanically and electrically connect to the high capacity removeable battery 502. The battery adaptor 504 is configured mechanically and electrically connect to the power tool carriage 108. When the battery adaptor 504 is mounted to the power tool carriage 108, the battery adaptor 504 is electrically connected to the power tool interface 204. In some other examples, the battery adaptor 504 is integral with the power tool carriage 108.

The battery adaptor 504 comprises an adaptor battery interface 702 configured to electrically and mechanically couple with the battery mounted tool interface 500 on the high capacity removeable battery 502.

The battery adaptor 504 in some examples is of a modular construction and is removably mountable to the power tool carriage 108. The battery adaptor 504 is mountable to the power tool carriage 108 and is fixed in place with one or more screw fastenings 700. This means that the battery adaptor 504 is fixed to the power tool carriage 108 during normal use. However, during manufacture or maintenance, the battery adaptor 504 can be replaced with another different battery adaptor 504. For example, the battery adaptor 504 as shown in FIG. 5 is suitable for e.g. a 60V high capacity removeable battery 502. In other examples, the battery adaptor 504 is configured to mechanically and electrically connect to any shape, size and capacity of battery. Another battery adaptor (not shown) can be replaced for the battery adaptor 504 as shown in FIG. 5 to connect a different battery to the power tool carriage 108. The different battery can have a different shape, size, voltage from the high capacity removeable battery 502 as shown in FIG. 5 or the removeable battery 130 as shown in FIG. 1.

The removeable battery adaptor 504 as shown in FIGS. 5 and 7 comprises a carriage interface (not shown) and the power tool carriage 108 comprises an adaptor interface (not shown) and carriage interface is compatible with the adaptor

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interface. This means that the carriage interface on the battery adaptor **504** and the adaptor interface on the power tool carriage **108** are arranged to electrically and mechanically connect together. This means that the battery adaptor **504** can be conveniently plugged into the power tool carriage **108** and makes manufacture easier.

In some examples, the carriage interface on the battery adaptor **504** is the same as the power tool interface **204** on the power tool carriage **108**. Furthermore in some examples the adaptor interface on the power tool carriage **108** is the same as the tool mounted battery interface **300** on the removable power tool **102**. In other words, when the battery adaptor **504** is removed from the power tool carriage **108**, the removable battery **130** as shown in FIG. 1 which is mountable on and compatible with the removable power tool **102** is mountable on the power tool carriage **108**.

When the tool mounted battery interface **300** on the removable power tool **102** is not compatible with the battery mounted tool interface **500**, this means that the different methods of attaching the high capacity removable battery **502** to power tool carriage **108** must be used compared to the method of attaching the removable power tool **102** to the power tool carriage **108**.

In some examples, the battery adaptor **504** comprises a latch mechanism **506**. The latch mechanism **506** comprises a spring biased hook portion (not shown) configured to engage a reciprocal recess in the high capacity removable battery **502**. The latch mechanism **506** secures the high capacity removable battery **502** to the battery adaptor **504**. The latch mechanism **506** is depressed to release the spring biased hook portion from the recess and the high capacity removable battery **502** is then removable from the battery adaptor **504**. Alternatively, the latch mechanism **506** can be mounted on the high capacity removable battery **502** and the reciprocal recess is located on the battery adaptor **504**.

Mounting the removable power tool **102** and the removable battery **130** to the power tool stand **100** will be discussed in more detail with reference to FIGS. 4, **6a** and **6b**. FIG. 4 shows the longitudinal axis A-A of the projecting frame **106**. Furthermore, FIG. 4 shows a first interface plane F-F aligned with a plane of the battery interface **206** mounted on the power tool carriage **108**. The removable battery **130** is mountable on the power tool carriage **108** in a direction E represented by the arrow. In some examples, the removable battery **130** is mountable in the direction E towards the base **104**. In some examples, the first interface plane F-F is substantially parallel with the longitudinal axis A-A of the projecting frame **106**.

FIG. 4 also shows a second interface plane H-H aligned with a plane of the power tool interface **204** mounted on the power tool carriage **108**. The removable power tool **102** is mountable on the power tool carriage **108** in a direction G represented by the arrow. In some examples, the removable power tool **102** is mountable in a direction G towards the base **104**. In some examples, the second interface plane H-H is substantially parallel with the longitudinal axis A-A of the projecting frame **106**. In some examples, the first and second interface planes are substantially parallel.

As shown in FIG. 4, the removable battery **130** and the removable power tool **102** are optionally mountable in the same direction E, G with respect to the power tool carriage **108**. This is advantageous because the user can use a similar pushing action in the same direction E, G to mount both the removable battery **130** and the removable power tool **102** on the power tool stand **100**. For example, the user can stand over the power tool stand **100** as shown in FIG. 1 and push down both the removable power tool **102** and the remove-

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able battery **130** to mount them on the power tool carriage **108**. This can make assembly of the removable power tool **102** and the removable battery **130** on the power tool carriage **108** easier and more intuitive.

In order to mount the removable battery **130** and the removable power tool **102** on the power tool carriage **108**, in some examples, both the removable battery **130** and the removable power tool **102** slide into mechanical engagement with the power tool carriage **108**. The sliding action of the removable battery **130** and the removable power tool **102** continues until a latch mechanism (not shown in FIG. 4) engages between the power tool carriage **108** and the removable battery **130** and/or the removable power tool **102**. In some examples, the battery interface **206** and the power tool interface **204** comprise two latch mechanisms which are the same as the latch mechanism used to hold the removable battery **130** in engagement when the mounted to the removable power tool **102**. The latch mechanism in some examples is the same as shown in reference to the latch mechanism **506** shown in FIG. 5 on the battery adaptor **504**.

The direction E, as shown in FIG. 4, is the direction that the removable battery **130** is mounted on the power tool carriage **108**. The direction E is in a direction opposite to the direction D that the removable battery **130** is mounted on the removable power tool **102**, as shown in FIG. 3.

Mounting the removable power tool **102** and the removable battery **130** on the power tool stand **100** will now be discussed in more detail in reference to FIGS. **6a**, and **6b**. FIGS. **6a** and **6b** respectively show a partial front and rear view of a power tool stand **100**.

FIGS. **6a** and **6b** respectively show the power tool interface **204** and the battery interface **206** without the removable power tool **102** or the removable battery **130** mounted thereon for the purposes of clarity.

The power tool interface **204** as shown in FIG. **6a** comprises a power tool mechanical interface **600** and a power tool electrical interface **602**. In some examples, the power tool mechanical interface **600** and the power tool electrical interface **602** are integral with the carriage housing **200**. For example, the power tool mechanical interface **600** and the power tool electrical interface **602** are moulded into the carriage housing **200**. In some examples, the power tool mechanical interface **600** and the power tool electrical interface **602** are positioned together or near each other on the carriage housing **200**. In some other less preferred examples, the power tool mechanical interface **600** and the power tool electrical interface **602** are located remote from each other on the carriage housing **200**.

As shown in FIG. **6a**, the power tool mechanical interface **600** optionally comprises a pair of rails **604**, **606** configured to engage with reciprocal slots mounted on the removable power tool **102**. The pair of rails **604**, **606** slide into engagement with the reciprocal slots and secure the removable power tool **102** to the power tool carriage **108**.

The power tool electrical interface **602** optionally comprises a plurality of electrical contact clips **608** configured to engage a plurality of reciprocal projecting contact ribs (not shown) mounted on the removable power tool **102**. For the purposes of clarity only one electrical contact clip **608** is labelled. The power tool mechanical interface **600** and the power tool electrical interface **602** are one optional way of connecting the removable power tool **102** to the power tool carriage **108**.

The battery interface **206** as shown in FIG. **6b** comprises a battery mechanical interface **610** and a battery electrical interface **612**. In some examples, the battery mechanical interface **610** and the battery electrical interface **612** are

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integral with the carriage housing 200. For example, the battery mechanical interface 610 and the battery electrical interface 612 are moulded into the carriage housing 200. In some examples, the battery mechanical interface 610 and battery electrical interface 612 are positioned together or near each other on the carriage housing 200. In some other less preferred examples, the battery mechanical interface 610 and the battery electrical interface 612 are located remote from each other on the carriage housing 200.

The battery mechanical interface 610 optionally comprises a pair of slots 614, 616 configured to engage with reciprocal rails mounted on the removeable battery 130. The battery electrical interface 612 comprises a plurality of projecting electrical contact ribs 618 configured to engage a plurality of reciprocal electrical contact clips (not shown) mounted on the removeable battery 130. For the purposes of clarity only one projecting electrical contact rib 618 is labelled in FIG.

6b. The battery mechanical interface 610 and the battery electrical interface 612 are one optional way of connecting the removeable power tool 102 to the power tool carriage 108.

In some examples, each of the power tool mechanical interface 600 and the battery mechanical interface 610 can optionally comprises a plurality of rails and slots to further improve the mechanical engagement when mounting the removeable power tool 102 and the removeable battery 130 on the power tool carriage 108. For example, each of the power tool mechanical interface 600 and the battery mechanical interface 610 can each comprise both slots and rails arranged to respectively engage with reciprocal rails and slots.

Alternatively, other mechanisms and features can be used for securely connecting the removeable power tool 102 to the power tool carriage 108. For example, in some less preferred alternatives, the rail and slot arrangement as discussed above is not used for mechanically securing the removeable battery 130 and the removeable power tool 102 to the power tool carriage 108. In this case, other external features on the removeable battery 130 and the removeable power tool 102 are used to securely fasten the removeable battery 130 and the removeable power tool 102 to the power tool carriage 108. In some examples, latches, straps, clips, clamps or any other suitable securing mechanism can be used to secure the removeable battery 130 and the removeable power tool 102.

As mentioned above, in some less preferred examples, the battery interface 206 comprises a battery mechanical interface 610 and a battery electrical interface 612 in separate locations on the power tool carriage 108. Similarly, the power tool interface 204 comprises a power tool mechanical interface 600 and a power tool electrical interface 602 in separate locations on the power tool carriage 108. This can be optionally achieved by making a separate electrical connection not using the previously discussed electrical contacts. For example, the battery electrical interface 612 and the power tool electrical interface 602 can be flexible ribbons projecting through the carriage housing 200 and the flexible ribbons have a plurality of electrical contacts connected thereto. The flexible ribbons can be manually attached to the ensure an electrical connection with the removeable power tool 102 and the removeable battery 130.

In another less preferred example, the removeable battery 130 can be mounted on the power tool carriage 108 in a different direction e.g. opposite or inclined to the direction that the removeable power tool 102 is mounted on the power tool carriage 108.

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The removeable power tool 102 is optionally mechanically connected to the power tool carriage 108 only at the power tool mechanical interface 600. However, in some other examples as shown in e.g. FIG. 2, the removeable power tool 102 is mounted to the power tool carriage 108 at a plurality of other mechanical couplings in addition to the power tool mechanical interface 600.

FIGS. 4, 6a and 6b discuss mounting the removeable battery 130 to the power tool carriage 108 and the removeable power tool 102 to the power tool carriage 108. However similar electrical and mechanical features discussed in reference to FIGS. 4, 6a and 6b can also be used to mount the high capacity removeable battery 502 to the battery adaptor 504. In addition similar electrical and mechanical features discussed in reference to FIGS. 4, 6a and 6b can be used to mount the battery adaptor 504 to the power tool carriage 108 in the example when the battery adaptor 504 is removably mountable to the power tool carriage 108.

In some other examples as shown in e.g. FIG. 1 and FIG. 2, the removeable power tool 102 is mounted to the power tool carriage 108 at a plurality of other mechanical couplings in addition to the power tool mechanical interface 600. Another mechanical coupling will now be discussed in reference to FIG. 2.

The power tool carriage 108 optionally comprises a tool clamp 214 configured to secure the removeable power tool 102 to the power tool stand 100. The tool clamp 214 is configured to engage a nose portion 126 (best shown in FIG. 1) of a housing 124 of the removeable power tool 102. The tool clamp 214 may optionally be secured to the nose portion 126 of the removeable power tool 102. When the tool clamp 214 is engaged with the nose portion 126, the removeable power tool 102 is further mechanically coupled to the power tool carriage 108. The coupling between the removeable power tool 102 and the power tool carriage 108 will be discussed in more detail below.

The tool clamp 214 optionally comprises a first clamp arm 216 and a second clamp arm 218 configured to urge towards each other. The first clamp arm 216 and the second clamp arm 218 each comprise a clamp jaw 220. The clamp jaws 220 on the first and second clamp arms 216, 218 are configured to engage a reciprocal housing recess in the nose portion 126 of the housing 124 of the removeable power tool 102. When the first and second clamp arms 216, 218 are engaged with the nose portion 126 between the first and second clamp arms 216, 218, the first and second clamp arms 216, 218 exert a gripping force against the nose portion 126.

By mounting the removeable power tool 102 to the power tool carriage 108 with the tool clamp 214, the removable power tool 102 does not move with respect to the power tool carriage 108. This means that the user can have the confidence that the removeable power tool 102 is securely mounted to the power tool stand 100 when performing a cutting operation.

As can be seen from FIG. 4, the removeable battery 130 is mounted on a first side 400 of the power tool carriage 108. The removeable power tool 102 is mounted on a second side 402 of the power tool carriage 108. The first side 400 and the second side 402 of the power tool carriage 108 are positioned either side of the projecting frame 106. Accordingly the first side 400 the second side 402 of the power tool carriage 108 are on opposite sides of the power tool carriage 108.

The power tool carriage 108 conveniently positions the removeable battery 130 remote from the removeable power tool 102 when both mounted on the power tool carriage 108.

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This means that the weight of the removeable power tool **102** and the removeable battery **130** can be better distributed about the projecting frame **106**. For example, the removeable power tool **102** can be mounted closer to the projecting frame **106**. This reduces the turning moment of the removeable power tool **102** when mounted on the power tool stand **100**. This makes the power tool stand **100** more stable when the removeable power tool **102** is mounted on the power tool carriage **108**.

By mounting the removeable battery **130** on an opposite side of the power tool carriage **108** on a different side of the projecting frame **106**, the weight of the removeable battery **130** provides a counterweight to the weight of the removeable power tool **102**. This further improves the stability of the power tool stand **100** when the removable power tool **102** and the removeable battery **130** are mounted on the power tool stand **100**.

The removeable battery **130** as shown in the accompanying Figs is only a single battery mounted on the power tool carriage **108**. However, in other examples, the power tool carriage **108** comprises a plurality of battery interfaces **204** mounted on the power tool carriage **108** configured to electrically and mechanically connected a plurality of removeable batteries **130**. This will be discussed in more detail below with respect to FIG. 8.

The carriage mechanism (not shown) is mounted within the carriage housing **200** and is arranged to move the power tool carriage **108** with respect to the projecting frame **106** when the carriage mechanism is actuated. The projecting frame **106** projects through a through hole **202** in the carriage housing **200**.

In some examples, the carriage mechanism comprises a rack (not shown) and pinion mechanism (not shown) for moving the power tool carriage **108** with respect to the projecting frame **106**. A rack is mounted on one side of the projecting frame **106** along the length of the projecting frame **106**. A drive gear is mounted on a drive shaft and a mechanism handle **128** is mounted on the drive shaft for manually rotating the drive shaft. The drive gear is optionally rotatably coupled to an intermediate gear and the intermediate gear engages the rack. In some other examples, the drive gear directly engages with the rack. In some other examples, any other suitable mechanism can be used to move the power tool carriage **108** with respect to the projecting frame **106**. In some other examples, the mechanism handle **128** is alternatively or additionally replaced with a drive mechanism coupled to a motor (not shown) for assisting the movement of the power tool carriage **108** on the power tool stand **100**.

Turning to FIG. 9, another example will now be described. FIG. 9 shows a side view of the power tool stand **100**. In some examples, the power tool stand **100** optionally comprises at least one accessory electrical interface **900** configured to connect to at least one accessory. The at least one accessory electrical interface **900** in some examples is an electrical socket **900**. The electrical socket **900** is shown in FIG. 9 as a standard mains power socket, however, the electrical socket **900** in some examples is a low voltage or a DC voltage socket. In some examples, the electrical socket **900** is adapted for non-mains power plugs so that only low voltage or DC voltage accessories can be inserted into the electrical socket **900**.

The electrical socket **900** is electrically connected to the battery interface **206** and/or the power tool interface **204**. By connecting the electrical socket **900** to the battery interface **206**, the removeable battery **130** can provide power to the

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electrical socket **900** when the removeable battery **130** is mounted to the battery interface **206**.

In some examples, the accessory that is connected to the electrical socket **900** is a water pump **1000** (best shown in FIG. 10) and/or a vacuum source **904** e.g. a vacuum pump **904** (best shown in FIG. 9). In other examples, the accessory can be any other suitable accessory such as a lamp, worksite radio or any other equipment used in the worksite.

The electrical socket **900** is optional and in some other examples is not provided. In this case, one or more accessories e.g. water pump **1000** (best shown in FIG. 10) and/or a vacuum source **904** e.g. a vacuum pump **904** (best shown in FIG. 9) are optionally powered from the removeable battery **130** via a power connection **922**. FIG. 9 schematically shows a power connection **922** between the removable battery **130** and the vacuum pump **904**. Similarly the water pump **1000** as shown in FIG. 10 can also be powered from the removeable battery **130** via the power connection **922** (but the power connection **922** is not shown in FIG. 10 for the purposes of clarity). Whilst FIG. 9 and FIG. 10 show different examples of the power tool stand **100** with different accessories (e.g. the vacuum pump **904** and the water pump **1000**) in another example, the power tool stand **1000** can comprise both the vacuum pump **904** and the water pump **1000**. In this case, both the vacuum pump **904** and the water pump **1000** can be powered from the removable battery **130** (or the high capacity removeable battery **502**).

Operation of the water pump **1000** and the vacuum pump **904** will be discussed below in more detail in reference to FIGS. 9 and 10 respectively.

In an alternative example, the accessory electrical interface **900** is optionally connectable to an external power source and the accessory electrical interface **900** is electrically connected to the first electrical interface and/or the second electrical interface. In some examples, the external power source is a mains power supply or an external battery. When a mains power supply is connected to the accessory electrical interface **900**, the mains power supply is arranged to power the removeable power tool **102** and/or charge the removeable battery **130** when the removeable power tool **102** and the removeable battery **130** are mounted on the power tool carriage **108**.

In some examples the switch **902** is an ON/OFF switch for the removable power tool **102**. In some examples, the switch **902** is in series with an ON/OFF switch e.g. the power trigger switch **308** on the removable power tool **102**. In this way, the switch **902** must be actuated before the removable power tool **102** is actuated. This reduces the chances of the removable power tool **102** being accidentally actuated when mounted on the power tool carriage **108**.

In some other examples, the switch **902** is alternatively an actuation switch configured to actuate the water pump **1000** independently of the removeable power tool **102**.

In some examples, the switch **902** is configured to wirelessly actuate the water pump **1000** or a vacuum pump **904**. In some examples, the switch **902** is configured to control a transmitter-receiver (not shown) mounted in the power tool carriage **108**. The transmitter-receiver is configured to send a receive signals from another transmitter-receiver mounted in the removable power tool **102** and/or the external accessory. The transmitter-receiver is configured to transmit one or more control signals for controlling the removeable power tool **102** and/or the external accessory. In this way, the control of the removeable power tool **102** can be remotely actuated by the power tool carriage **108** via transmission of actuation signals. In some examples the transmitter-receiver mounted on the power tool carriage **108** is configured to

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transmit actuation signals directly to a transmitter-receiver mounted in the external accessory. Additionally or alternatively transmitter-receiver mounted on the power tool carriage **108** is configured to transmit actuation signals indirectly to a transmitter-receiver mounted in the external accessory via a transmitter-receiver mounted in the removable power tool **102**. In this way, the switch **902** can send a signal to switch off the vacuum pump **904** and/or the water pump **1000**.

Water Pump

A problem with some removable power tools **102** e.g. diamond core drills is that they need suitable lubrication management during operation. The power tool stand **100** as shown in the Figs. comprises a suitable lubrication management which will now be discussed in more detail.

As mentioned above, the power tool carriage **108** comprises at least one duct **222** configured to convey a fluid. The at least one duct **222** is best shown in FIG. 2. The at least one duct **222** in some examples is configured to supply a lubrication fluid such as water to the removable power tool **102**. Accordingly, the at least one duct **222** is in fluid communication with a lubrication fluid supply e.g. a water supply such as a water pump **1000** or a water tap (not shown). This allows the at least one duct **222** to supply water to the removable power tool **102** during use.

In some other examples, the at least one duct **222** is in fluid communication with an air supply or a vacuum source (not shown). Accordingly, the at least one duct **222** is in fluid communication with an air supply or a vacuum source e.g. a compressed air supply (not shown) or a vacuum pump (not shown) configured to extract dirt and debris. This allows the at least one duct **222** to extract debris from the removable power tool **102**.

The at least one duct **222** can be optionally used for conveying a fluid e.g. a lubrication fluid such as water towards the removable power tool **102** or conveying a fluid e.g. an airflow away from the removable power tool **102**. Hereinafter, the at least one duct **222** will be referred to as a water duct **222**, but this could also be an air duct as previously discussed.

As shown in FIG. 2 the water duct **222** is mounted on the outside of the power tool carriage **108**. The water duct **222** extends along the surface of the power tool carriage **108**. Although not shown, in some examples, the water duct **222** can be secured to the power tool carriage **108** with clips, clamps, or adhesive or any other suitable means. In some alternative examples which are shown in e.g. FIG. 4, the water duct **222** is mounted inside the power tool carriage **108**.

The water duct **222** comprises a duct inlet **404** in fluid communication with the water supply. In some examples, the duct inlet **404** comprises a toolless connector for connecting to a water hose. The toolless connector allows for a snap fit connection between the duct inlet **404** and a water hose (not shown). The water duct **222** is also in fluid communication with a fluid connector **224**. The water duct **222** can supply water to the moveable fluid connector **224**. The fluid connector **224** is optionally moveable between a first position and a second position. FIG. 2. shows the moveable fluid connector **224** in the first position. In the second position the fluid connector **224** is engaged with the removable power tool **102**.

The fluid connector **224** is mounted a connector housing **226**. The connector housing **226** protects the fluid connector

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224 and houses a fluid connection between the water duct **222** and the fluid connector **224**.

Alternatively, or additionally the fluid connector **224** is connectable with an accessory such as a nozzle (not shown). The optional nozzle is arranged to eject a spray of water on the cutting tool **132** in order to cool and lubricate the cutting tool **132**. In some other examples, the fluid connector **224** is connectable to an optional adapter (not shown) for guiding the flow of water around the cutting tool **132** and/or the removable power tool **102**.

In some examples, the water duct **222** comprises a carriage valve (not shown) arranged to selectively adjust the flow of fluid to and/or from the portion of the removable power tool **102**. The carriage valve is arranged to be adjustable by the user to selectively adjust the flow of water to the cutting tool **132**. The user can optionally fully open or fully close the carriage valve in order to actuate the flow of water through the water duct **222**. Furthermore, the carriage valve can optionally be positioned at any position between a fully closed and a fully opened state. This means that the user can adjust the flow rate by manipulating the carriage valve on the power tool carriage **108** depending on the amount of lubrication required during a cutting operation.

As mentioned above, in some examples, the power tool stand **100** comprises a water pump **1000** as shown in FIG. 10. The water pump **1000** as shown in FIG. 10 is mounted inside the carriage housing **200** of the power tool carriage **108**. Alternatively, the water pump **1000** can be located outside of the power tool carriage **108**. The water pump **1000** can be mounted to the outside of the power tool carriage **108**. Alternatively, the water pump **1000** can be mounted to the base **104**.

The water pump **1000** is electrically connected to the removable battery **130** when the removable battery **130** is mounted to the power tool carriage **108**. This means that the removable battery **130** is configured to power to the water pump **1000**. In some examples the power tool carriage **108** comprises a water pump control switch **1002** connected in series between the water pump **1000** and the removable battery **130**. The user can then manually turn on the water pump **1000** when needed by actuating the water pump control switch **1002**. In some examples, the controller **1100** (best shown in FIG. 11 or 14) can selectively control actuation of the water pump **1000**. For example, the controller **1100** can issue a control signal to the water pump **1000** to actuate the water pump **1000** when the user actuates the power trigger switch **308** on the removable power tool **102**.

The water pump **1000** is in fluid communication with a water reservoir **1004**. The water reservoir **1004** is optionally a bucket of water or a water tank. The water pump **1000** can be connected to the water reservoir **1004** via a water hose **1006**. The water reservoir **1004** is unpressurised. The water pump **1000** can be any suitable pump configured to draw water from the water reservoir **1004** and convey the water to the water duct **222**.

In some examples, the removable battery **130** is configured to be in electrical connection with both the removable power tool **102** and the water pump **100** when the removable battery **130** and the removable power tool **102** are mounted to the power tool carriage **108**.

In some other examples, the removable battery **130** mounted to power tool carriage **108** is in electrical connection with the water pump **1000** but not the removable power tool **102**. In this case, the power tool carriage **108** comprises a power tool interface which mechanically couples the removable power tool **102** to the power tool carriage **108**.

However, when the removeable power tool **102** and the removeable battery **130** are mounted on the power tool carriage **108**, the removeable power tool **102** and the removeable battery **130** are in different electrical circuits. This means that the removeable battery **130** is only in electrical connection with the water pump **1000**. In this example, the removeable power tool **102** is powered with an alternate power source e.g. another removeable battery or a mains power supply etc.

In another example, the removeable power tool **102** is electrically connected to a first removeable battery **130** and the water pump **1000** is electrically connected to a second removeable battery **800**. Similarly the first removeable battery **130** and the second removeable battery **800** are not electrically connected together and are in different electrical circuits. The first removeable battery **130** and the second removeable battery **800** will be discussed in more detail below.

Dual Battery and Battery Control

As mentioned above, in some examples the power tool stand **100** is configured to mechanically and electrically connect to a plurality of batteries **130**, **800**. For example, a first removeable battery **130** and a second removeable battery **800** can be mounted on the power tool carriage **108** side by side as shown in FIG. **8**. The power tool carriage **108** comprises a first battery interface **802** and a second battery interface **804** configured to respectively electrically and mechanically connect the first and second removeable batteries **130**, **800** to the power tool carriage **108**. The first battery interface **802** and the second battery interface **804** are identical to the battery interface **206** discussed above and shown in e.g. FIG. **2**. Alternatively, any number of removeable batteries **130** can be mounted on a similar number of battery interfaces **204**.

Any of the other examples can be used together with the arrangement discussed in reference to FIG. **8**. For example, the examples using the water pump **1000** as shown in FIG. **10** or the examples using the vacuum pump **904** as shown in FIG. **9** can be used together with the examples discussed in reference to FIG. **8**.

In some examples, each of the first removeable battery **130** and the second removeable battery **800** are mounted to the power tool carriage **108** with the battery adaptor **504** as shown in FIGS. **5** and **7**. In other words, each of the first battery adaptor **806** and the second battery adaptor **808** as shown in FIG. **8** is the same as the battery adaptor **504** as discussed in reference to FIGS. **5** and **7**.

Examples described with reference to FIGS. **8**, **14** and **15** discuss the control of the first battery **130** and the second battery **800**. FIG. **14** shows a schematic view of the power tool stand **100** with the removeable power tool **102**, the first battery **130** and the second battery **800** mounted to the power tool carriage **108**. FIG. **15** shows a flow diagram of operation of a power tool stand **100**.

The first battery **130** and the second battery **800** can be each a single battery pack. In other examples, each of the first battery **130** and the second battery **800** can be each a group of battery packs or a group of battery cells. In some examples there can be other battery packs. There can be any number of other battery packs as required. The example discussed below will only have a first battery **130** and a second battery **800** however the control functionality can be applied to any number of battery packs connected to the circuit **1400** of the power tool stand **100**. The first and second batteries **130**, **800** are identical to the removeable

battery **130** as shown in FIG. **1** or alternatively identical to the high capacity removeable battery **502** as shown in FIG. **5**.

The first and second batteries **130**, **800** are electrically connectable to the removeable power tool **102** when mounted on the power tool carriage **108**. The first and second batteries **130**, **800** can be connected together to the removeable power tool **102** in series or parallel. Alternatively, first and second batteries **130**, **800** are connectable separately to the removeable power tool **102**. In other words, only one of the first and second batteries **130**, **800** is connected to the removeable power tool **102** at any one time.

In this example, either a controller **1100** or a user is configured to selectively connect the first battery **130** and/or the second battery **800** to the removeable power tool **102**. A user can selectively connect the first battery **130** and/or the second battery **800** to the removeable power tool **102** by pressing a user actuated switch **1402** on the power tool stand **100**. The user actuated switch **1402** may change the connection between the first battery **130** to the removeable power tool **102** to a connection between the second battery **800** and the removeable power tool **102**.

In some examples, the controller **1100** can switch the connection of first and second batteries **130**, **800** to the removeable power tool **102** between a series connection of the first and second batteries **130**, **800**, a parallel connection of the first and second batteries **130**, **800**, a single connection of the first battery **130** and a single connection of the second battery **800**.

The controller **1100** is connected to the first battery **130** and the second battery **800**. The controller **1100** is configured to issue control instructions to the first and second batteries **130**, **800**. In some examples, the controller **1100** is a separate battery controller configured to control the connections of the first and second batteries **130**, **800** to the removeable power tool **102**.

However, in other examples the controller **1100** is mounted within the power tool carriage **108** and is configured to control the connections of the first and second batteries **130**, **800**. Accordingly the controller **1100** is configured to selectively connect the first battery **130** and/or the second battery **800** to the removeable power tool **102**.

The controller **1100** is configured to receive signals relating to parameters and/or variables for the first and second batteries **130**, **800** as shown in step **1500**. In a first example, the controller **1100** optionally receives a manual input from a user. The manual input from the user is received from a user actuated battery switch **1402** mounted on the power tool carriage **108**. For example, the user actuated battery switch **1402** can be positioned next to the water pump control switch **1002** as shown in FIG. **10**. When the controller **1100** receives the signal from the user actuated switch **1402**, the controller **1100** determines a status of the first and second batteries **130**, **800** as shown in step **1502**. For example, the actuation of the battery switch **1402** indicates to the controller **1100** that the first battery **130** connected to the removeable power tool **102** has a low charge status.

The user may actuate the battery switch **1402** in response to the user seeing that the charge in the first battery **130** is depleted. The user may determine the charge status from the first battery **130** itself. For example, the first battery **130** may comprise charge indication LEDs (not shown). In another example, a display (e.g., a row of LEDs or a screen) may be provided on the power tool carriage **108** (or any other part of the power tool stand **100** that is in the line of sight of the user from the operating position) that displays a state of charge for each of the first and second batteries **130**, **800**.

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The display may be electrically connected to first and second batteries 130, 800 directly or via the controller 1100. In another example, the display may wirelessly communicate (e.g., via Bluetooth) with the controller 1100 to receive state of charge information wirelessly.

Once the controller 1100 determines that the user has actuated the battery switch 1402, the controller 1100 determines that a change in status of the first battery 130 has occurred as shown in step 1504. This means that there is a required change connection status of the current battery e.g. the first battery 130 connected to the removeable power tool 102. The controller 1100 then issues a control instruction to the circuit 1400 to connect the second battery 800 to the removeable power tool 102 as shown in step 1506. At the same time the controller 1100 issues another control instruction to disconnect the first battery 130 from the removeable power tool 102. In this way the controller 1100 separately connects the first or second batteries 130, 800 to the removeable power tool 102 as shown in step 1508. Once the second battery 800 is connected to the removable power tool 102, the user can continue operating the removable power tool 102.

In another example, there is optionally no manual input from the user via the battery switch 1402. Instead the controller 1100 selectively controls the connection of the first battery 130 and/or the second battery 800 automatically. For example, the controller 1100 can switch the connection to the removeable power tool 102 between the first and second batteries 130, 800 in dependence of the operational status of the first and second batteries 130, 800.

In some examples, in step 1500 the controller 1100 is configured to receive operational status information from the first and second batteries 130, 800. The operational status information can comprise one or more of the following battery voltage, battery capacity, battery temperature, battery discharge rate, battery current or any other battery parameter or status information. In some examples, the controller 1100 is connected to a voltage sensor 1404 and a current sensor 1406 to respectively send voltage signals and current signals to the controller 1100.

Once the controller 1100 has received the signal from the first and second batteries 130, 800, the controller 1100 determines the current status information of the first and second batteries 130, 800 in step 1502. For example, the controller 1100 determines that the first battery 130 which is connected to the removeable power tool 102 has an instantaneous operating temperature, an instantaneous operating voltage and/or an instantaneous remaining battery capacity.

The controller 1100 then compares the determined status information of the first and second batteries 130, 800 to predetermined threshold values as shown in step 1510. The predetermined threshold values may optionally comprise operating voltage ranges, operating battery capacity and operating battery temperature. In some examples, the predetermined threshold values are a look up table stored in memory of the controller 1100.

For example, in step 1510 the controller 1100 determines that the battery capacity of the first battery 130 connected to the removeable power tool 102 has dropped below a threshold of e.g. 30%, 25%, 20%, 15%, 10%, or 5% of the total battery capacity of the first battery 130.

Alternatively, in step 1510 the controller 1100 determines that the operating temperature of the first battery 130 connected to the removeable power tool 102 has exceeded an operating temperature threshold e.g. 60° C., 55° C., 50° C., 45° C., 40° C. etc.

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Once the controller 1100 has determined that the first battery 130 is operating outside the normal operating parameters in step 1510, the controller 1100 determines similarly in step 1504 that the operational status of the first battery 130 has changed. Accordingly, the controller 1100 issues a control instruction in step 1506 to the circuit 1400 to connect the second battery 800 to the removeable power tool 102 and disconnect the first battery 130 from the removeable power tool 102 as shown in step 1508.

If the controller 1100 determines that the first battery 130 is operating normally in step 1504 and there has not been a change in the operational status of the first battery 130, the controller 1100 does not issue a control instruction to change the connection to the removeable power tool 102 between the first and second batteries 130, 800. Instead, the controller 1100 returns to step 1500 continues to receive signals relating to parameters and/or variables relating to the first and second batteries 130, 800.

Optionally, the controller 1100 can issue a control instruction to the circuit 1400 to connect the first and second batteries 130, 800 in parallel as shown in step 1512 or in series as shown in step 1514. This can provide a different voltage and current to the removeable power tool 102 if required.

As mentioned above, in some examples the first and second batteries 130, 800, comprise battery circuits comprising a battery controller. The controller 1100 in some examples is configured to receive operational data from the battery controller as shown in step 1500. In some examples, the operational data is one or more error signals e.g. an error code indicating a fault or malfunction of the first or second batteries.

In this case in the controller 1100 determines a fault status of the first battery 130 in step 1502. The controller 1100 determines that the first battery 130 has changed from normally operating to malfunctioning and needs to be disconnected from the removeable power tool 102. The controller 1100 then connects the second battery 800 to the removeable power tool 102 and disconnects the first battery 130 as previously discussed.

In another example, the controller 1100 is optionally configured to optimizing the voltage provided to the removeable power tool 102 by selectively connecting the first and second batteries 130, 800 to the removeable power tool 102. In this way, the controller 1100 is configured to optimize the voltage provided to the removeable power tool 102 as the first and second batteries 130, 800 discharge.

The controller 1100 is optionally configured to determine a removable power tool 102 parameter e.g. a motor parameter in step 1516. The controller 1100 determines that the motor of the removeable power tool 102 may be rated for a first voltage, e.g., 40V. The controller 1100 in some examples determines the motor voltage rating based on information stored in memory of the controller 1100. The controller 1100 may determine that the first and second batteries 130, 800, at full charge, provide a second voltage. In some examples, the second voltage is higher than the first voltage e.g., 60V.

In some cases, the controller 1100 selectively connects the first and second batteries 130, 800 to the removeable power tool 102. This means that the controller 1100 does not need to use pulse width modulation (PWM) at a duty cycle to provide the lower first voltage to the removeable power tool 102 from e.g., the first battery 130 at the higher second voltage.

This can be advantageous because the controller 1100 can avoid the inherent losses caused by PWM switching. In this

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case, the controller 1100 can selectively connect the first and second batteries 130, 800 to the removeable power tool 102 with a 100% duty cycle.

In some examples, in steps 1502, 1504 the controller 1100 determines the current battery capacity of the first battery 130 and the second battery 800. The controller 1100 may determine that the first battery 130 and the second battery 800 may have a different state of charge. Accordingly, the controller 1100 determines that the first battery 130 and the second battery 800 have different instantaneous voltages.

After the controller 1100 determines the motor voltage rating in step 1516, the controller 1100 determines which of the first and second batteries 130, 800 has an instantaneous voltage closer to the motor voltage rating. The controller 1100 in step 1506 connects to either the first battery 130, or the second battery 800 in dependence of which instantaneous voltage of the first and second batteries 130, 800 is closer to the motor voltage rating. In this way, the controller 1100 can supply the correct voltage to the motor of the removeable power tool 102 whilst minimizing the PWM switching.

In step 1506, if the controller 1100 determines that the motor voltage rating is 40V and the first battery 130 has an instantaneous voltage of 60V and the second battery 800 has an instantaneous voltage of 45V, then the controller 1100 will connect the second battery 800 to the motor of the removeable power tool 102. This is because the voltage of the second battery 800 is closer to the motor voltage rating. This will mean that it may be more efficient to run the motor of the removeable power tool 102 from the second battery 800 so that limited or no PWM switching is needed.

In another example, controller 1100 determines that the motor of the removeable power tool 102 requires different voltages in dependence of the motor speed. Accordingly in step 1516, the controller 1100 determines the required motor speed. Information relating to the required motor speed in some examples is received by the controller 1100 in a signal. The signal may be generated by a user input e.g., a trigger switch or a speed dial. In step 1516, the controller 1100 determines that a high motor speed or a low motor speed is required. The controller 1100 determines the high motor speed voltage or the low motor speed voltage. Accordingly, the controller 1100 determines the required motor voltage. The controller 1100 determines in step 1506 which of the voltages of the first or second batteries 130, 800 most closely matches the required motor voltage.

For example, the controller 1100 may determine that the required motor voltage for a low-speed rating is 40V and the first battery 130 has an instantaneous voltage of 60V and the second battery 800 has an instantaneous voltage of 40V. In this case, the controller 1100 determines in step 1506 that the second battery 800 voltage is closest to the required motor voltage. The controller 1100 then connects the second battery 800 to the motor of the removeable power tool 102.

Alternatively, the controller 1100 may determine that the required motor voltage for a high-speed rating is 60V and the first battery 130 has an instantaneous voltage of 60V and the second battery 800 has an instantaneous voltage of 40V. In this case, the controller 1100 determines in step 1506 that the first battery 130 voltage is closest to the required motor voltage. The controller 1100 then connects the first battery 130 to the motor of the removeable power tool 102.

Furthermore, in some examples, the controller 1100 is configured to connect the first and second batteries 130, 800 in series in step 1514, or in parallel in step 1512, in order to

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generate a voltage supplied by the first and second batteries 130, 800 as close as possible to match the required motor voltage.

Vacuum Seal

Another example will now be discussed in reference to FIGS. 9, 11, 12, 13 and 14. FIG. 9 shows a side view of the power tool stand 100. FIG. 11 shows a schematic view of the power tool stand 100. FIG. 12 shows a graph of pressure of a workpiece surface seal 906 time. FIG. 13 shows a flow diagram of operation of the power tool stand 100.

FIG. 14 shows a schematic view of the power tool stand 100.

The power tool stand 100 as shown in FIG. 9 is the same as discussed in reference to the FIGS. 1 to 8 except that the base 104 is modified. The base 104 comprises a work surface seal 906 mounted in the base 104 and configured to engage the workpiece surface 920. The work surface seal 906 is configured to maintain a vacuum seal against the workpiece surface 920. This means that the work surface seal 906 provides a suction force for fixing the power tool stand 100 against the workpiece surface 920. The workpiece surface 920 in some examples (not shown in FIG. 9) is a vertical surface such as a wall. This means that the work surface seal 906 allows the power tool stand 100 to be more easily or better fixed to the vertical surface.

Whilst FIG. 9 shows a single work surface seal 906, in other examples there can be a plurality of work surface seals 906 mounted on the base 104.

In some examples, the work surface seal 906 generates a sufficiently large suction force that only the work surface seal 906 is needed to fix the power tool stand 100 to the workpiece surface 920. In other examples, the work surface seal 906 is used together with other fixing attachments e.g. the fixing bolts 120 as shown in FIG. 1.

The work surface seal 906 is in fluid communication with a vacuum source 904. In some examples the vacuum source 904 is a vacuum pump 904 connected to the work surface seal 906 via a hose 908. Whilst FIG. 9 shows the vacuum pump 904 remote from the power tool stand 100, in some other examples, the vacuum pump 904 is mounted on the power tool carriage 108 or the base 104. The vacuum pump 904 as shown in FIG. 9 is optionally mounted in a vacuum pump housing which is mounted to the base 104. The vacuum pump housing can be fixed to base 104 via threaded fasteners. One or more suction conduits or hoses are in fluid communication between the vacuum pump 904 and the work surface seal 906. The one or more suction conduits or hoses can extend through the base 104 to the work surface seal 906 as schematically shown in FIG. 9.

Alternatively, the vacuum pump 904 is mounted within the power tool carriage 108. The vacuum pump 904 comprises a motor fan assembly 910 configured to generate a negative pressure airflow. When the vacuum pump 904 is actuated, the vacuum pump 904 is configured to generate a partial pressure or vacuum pressure in the work surface seal 906 which causes a suction force on the work surface seal 906 against the workpiece surface 920.

Any suitable vacuum source 904 means can be used to generate to the negative pressure air flow to cause the suction force. For example, the arrangement as shown in FIG. 9 shows a vacuum source 904 connected via the hose 908 to the work surface seal 906. Alternatively, the negative pressure air flow can be generated next to the work surface seal 906 in the base 104. For example, a venturi nozzle (not shown) mounted next to and in fluid communication with

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the work surface seal **906** can be connected to a source of compressed air. The venturi nozzle can generate the negative pressure air flow when the source of compressed air flows into the venturi nozzle. The vacuum pump **904** can be manually actuated by a user operated vacuum pump control switch **1106**. This means that the control of the vacuum pump **904** can be achieved manually and without the need of a controller **1100** or a pressure sensor **912** as shown in FIG. 9. In some examples, once the user manually actuates the vacuum pump **904**, the vacuum pump **904** is configured to generate a vacuum pressure at a predetermined pressure e.g. a pressure set point. This means that the suction force can be applied at the work surface seal **906** without automatic control.

The vacuum pump **904** is optionally electrically connected to the removeable battery **130** when the removeable battery **130** is mounted to the power tool carriage **108**. This means that the removeable battery **130** is configured to power to the vacuum pump **904**. In some examples the power tool carriage **108** comprises a vacuum pump control switch **1106** connected in series between the vacuum pump **904** and the removeable battery **130**. The user can then manually turn on the vacuum pump **904** when needed by actuating the vacuum pump control switch **1106**. In some other examples, the vacuum pump **904** is alternatively optionally electrically connected to a separate independent battery (not shown) mounted to the vacuum pump **904** or the power tool stand **100**. In this case, there are two removable batteries **130** mounted to the power tool stand which are each part of a separate electrical circuit e.g. an electrical circuit for the removeable power tool **102** when mounted on the power tool stand **100** and another electrical circuit for the vacuum pump **904**.

Additionally or alternatively, the vacuum pump **904** can be optionally controlled by a controller **1100** and vacuum source **904** is connected to a controller **1100**. Alternatively, this means that the vacuum pump **904** can be controlled without manual input from the user as mentioned above. The controller **1100** is configured to issue control instructions to the vacuum source **904** and/or the removeable power tool **102**. For example, the controller **1100** is configured to actuate and control the speed and direction of the motor fan assembly **910**. The controller **1100** is also configured to control the removeable power tool **102** in response to one or more detected conditions in the work surface seal **906**. The controller **1100** is connected to the removeable power tool **102** via a first data connection **914**. The controller **1100** as shown in FIG. 9 is external to the power tool stand **100**. However, in some other examples, the controller **1100** is mounted within the power tool carriage **108**.

The controller **1100** is also optionally connected to at least one sensor **912** via a second data connection **916**. The at least one sensor **912** is a pressure sensor **912** configured to detect the pressure in the work surface seal **906**. The pressure sensor **912** is configured to send a pressure signal to the controller **1100**. The pressure sensor **912** can periodically detect the pressure and send a pressure signal to the controller **1100**. In other examples, the pressure sensor **912** constantly sends a pressure signal to the controller **1100**. In some examples, the pressure sensor **912** is configured to detect the vacuum pressure at the work surface seal **906**.

In some other examples, the controller **1100** is configured to receive the raw pressure signal from the pressure sensor **912** and determine the vacuum pressure at the work surface seal **906**.

The controller **1100** in some examples is optionally connected to a vacuum pump voltage sensor **1102** and a vacuum

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pump current sensor **1104** which are configured to respectively send voltage and current signals relating to the motor fan assembly **910** of the vacuum source **904**. This means that the controller **1100** can optionally determine the load on the vacuum source **904** and determine whether there is a fault with the vacuum source **904** or whether an adequate seal has been made between the work surface seal **906** and the workpiece surface **920**.

Operation of the work surface seal **906** will now be discussed in further detail with respect to the FIG. 13.

The controller **1100** receives a pressure signal from the pressure sensor **912** in step **1300**. The controller **1100** then determines whether the received pressure signal is below a first vacuum pressure threshold **1200** as shown in step **1302**. At $T=T1$ as shown in FIG. 12 the controller **1100** determines that the received pressure signal is at a first vacuum pressure signal. It should be noted that the graph in FIG. 12 shows the vacuum pressure at the work surface seal **906** over time. This means a higher vacuum pressure indicates a lower absolute pressure and a lower vacuum pressure indicates a higher absolute pressure. In other words as the vacuum pressure tends to zero in FIG. 12, the absolute pressure in the work surface seal **906** tends to atmospheric pressure.

The first vacuum pressure threshold **1200** is a threshold that indicates that the seal between the work surface seal **906** and the workpiece surface **920** is not perfect. This means that there is a suction force being exerted by the work surface seal **906**, but this is not a maximum suction force that can be exerted by the work surface seal **906**.

When the vacuum pressure in the work surface seal **906** is between the first vacuum pressure threshold **1200** and a second vacuum pressure threshold **1202**, the power tool stand **100** and removeable power tool **102** are operable. However, the user should be notified that suction force may not be sufficient to safely operate the removeable power tool **102** on the power tool stand **100**. The second vacuum pressure threshold **1202** is lower than the first vacuum pressure threshold **1200**. This means that a pressure in the work surface seal **906** at the second vacuum pressure threshold **1202** has a worse seal between the work surface seal **906** and the workpiece surface **920** than when the pressure in the work surface seal **906** is at the first vacuum pressure threshold **1200**.

Since the controller **1100** determines that the pressure is above the first vacuum pressure threshold **1200**, the controller **1100** returns to step **1300** and continues monitoring the pressure in the work surface seal **906**.

At time $T=T2$, the controller **1100** determines that the pressure in the work surface seal **906** has dropped below the first vacuum pressure threshold **1200**. This means that the vacuum pressure in the work surface seal **906** is dropping. The controller **1100** then determines whether the pressure in the work surface seal **906** has dropped below a second vacuum pressure threshold **1202** in step **1304**. At $T=T2$, the controller **1100** determines that the pressure is still above the second vacuum pressure threshold **1202**.

The controller **1100** then issues an alert to the user in step **1306**. The alert can be displayed on an LCD display (not shown) of the removeable power tool **102** or on the power tool stand **100**. Alternatively, the alert can be illuminating an LED on the removeable power tool **102** or the power tool stand **100**. Alternatively, the alert can be an audible signal or a haptic signal.

If the controller **1100** determines whether the pressure in the work surface seal **906** has dropped below a second vacuum pressure threshold **1202** in step **1304**, e.g. at $T=T3$, the controller **1100** can issue a control instruction to stop the

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removeable power tool **102** as shown in step **1308**. The controller **1100** may actuate a switch to cut the power to the removeable power tool **102** in response to determining that the vacuum pressure at the work surface seal **906** has fallen below the second vacuum pressure threshold **1202**. In this case, the controller **1100** may stop the vacuum source **904** as well. Accordingly the vacuum pressure falls to zero at $T=T4$.

This means that the user can be protected from the power tool stand **100** detaching from the workpiece surface **920** and falling on the user.

In some examples, the controller **1100** in step **1310** may optionally issue a control instruction to the vacuum source **904** following a determination that the vacuum pressure is falling in the work surface seal **906** at $T=T5$. The controller **1100** issues a control instruction to increase the speed of the motor fan assembly **910** and increase the negative pressure air flow. This can increase the vacuum pressure in the work surface seal **906** such that the vacuum pressure exceeds the first vacuum pressure threshold **1200** at $T=T6$. The controller **1100** can control the pressure in the work surface seal **906** with a feedback loop from the pressure sensor **912** and controlling the vacuum source **904** as shown in FIG. **13**.

Advantageously, by using the pressure sensor **912** which detects the vacuum pressure in the work surface seal **906**, the controller **1100** can directly determine whether the vacuum seal against the workpiece surface **920** has deteriorated more accurately. This means the controller **1100** can inform the user more accurately and safely in case there is an issue with the work surface seal **906**.

In another example, two or more examples or features are combined. Features of one example can be combined with features of other examples e.g. any of the features described above in an example or shown in one or more Figs can be combined with one or more other features described above or shown in one or more Figs. Examples of the present disclosure have been discussed with particular reference to the examples illustrated. However it will be appreciated that variations and modifications may be made to the examples described within the scope of the disclosure.

The invention claimed is:

1. A power tool stand for a removeable power tool having a first battery interface and a removeable battery having a first tool interface, the power tool stand comprising:

a base;
a projecting frame connected to the base; and
a power tool carriage moveably mounted on the projecting frame, the power tool carriage comprising a second battery interface and a second tool interface;
wherein the second battery interface is compatible with the first tool interface and the second battery interface is arranged to electrically and mechanically connect together with the first tool interface; and
the second tool interface is compatible with the first battery interface and the second tool interface is arranged to electrically and mechanically connect together with first battery interface;
and the first battery interface and the first tool interface are incompatible.

2. A power tool stand according to claim 1, wherein the power tool carriage comprises a battery adapter comprising the second battery interface.

3. A power tool stand according to claim 2, wherein the battery adapter is mountable to the power tool carriage.

4. A power tool stand according to claim 3, wherein the battery adaptor is removably mountable to the power tool carriage.

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5. A power tool stand according to claim 2, wherein the battery adapter comprises a carriage interface and the power tool carriage comprises an adaptor interface and adaptor interface is compatible with the carriage interface and the adaptor interface is arranged to electrically and mechanically connect together with carriage interface.

6. A power tool stand according to claim 5, wherein the carriage interface is the same as the second tool interface and the adaptor interface is the same as the first battery interface.

7. A power tool stand according to claim 1, wherein the second battery interface is configured to mechanically and electrically connect to the removable battery, and the removable battery has a voltage of at least 60V.

8. A power tool stand according to claim 1, wherein the power tool stand comprises a rack and pinion mechanism arranged to move the power tool carriage relative to the projecting frame.

9. A power tool stand according to claim 1, wherein the power tool carriage comprises a handle coupled to the rack and pinion mechanism.

10. A power tool stand according to claim 1, wherein the second battery interface is mounted on a first side of the power tool carriage and the second tool interface is mounted on a second side of the power tool carriage.

11. A power tool stand according to claim 1, wherein the second tool interface projects over the base.

12. A power tool stand according to claim 1, wherein the power tool stand comprises a plurality of the second battery interfaces arranged to electrically connect with a plurality of removeable batteries.

13. A power tool stand according to claim 1, wherein the power tool carriage comprises a latch recess for engaging with a latch mechanism mounted on the removable battery.

14. A power tool stand according to claim 1, wherein the power tool carriage comprises an air duct connectable to a dust extractor.

15. A power tool stand according to claim 1, wherein the power tool carriage comprises a water duct connectable to a water supply.

16. A power tool stand according to claim 1, wherein the second battery interface is electrically connected to the second tool interface within the housing of the power tool carriage.

17. A power tool stand system comprising:

a power tool stand, the power tool stand comprising:
a base;
a projecting frame connected to the base; and
a power tool carriage moveably mounted on the projecting frame, the power tool carriage comprising a second battery interface and a second tool interface;
a removeable power tool having a first battery interface, wherein the first battery interface is connectable to the second tool interface of the power tool stand; and
a first removable battery having a first tool interface, wherein the first tool interface is connectable to the second battery interface of the power tool stand.

18. A power tool stand system according to claim 17, further comprising a second battery having a third tool interface compatible with the first battery interface.

19. A power tool stand, comprising:

a base;
a projecting frame; and
a power tool carriage configured to support a removable power tool and to translate relative to the projecting frame, the power tool carriage comprising a second battery interface and a second tool interface;

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wherein the second battery interface is configured to electrically and mechanically connect together with a first tool interface of a removable battery pack;

wherein the second tool interface is configured to electrically and mechanically connect together with a first battery interface of the removable power tool; and wherein the first battery interface and the first tool interface are incompatible.

20. The power tool stand of claim **19**, wherein the second battery interface is at a first side of the power tool carriage and the second tool interface is at a second side of the power tool carriage.

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